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A Project Report On

Apartment Management System

AN WEB APPLICATION TO HELP HOUSE OWNER & RENTER

Course Code: CSE-4800

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A project submitted in partial fulfillment of the requirements for the Degree of
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CERTIFICATE OF APPROVAL

The project entitled as “Apartment Management System” submitted by Shahida Sultana, bearing ID No: C161202 and Ishrat Jahan, bearing ID No: C161219 of the Department of Computer Science & Engineering (CSE) of International Islamic University Chittagong (IIUC) has been accepted as satisfactory for the partial fulfillment of the requirements for the Degree of Bachelor in Computer Science & Engineering.

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DECLARATION

Where by declare that the work has been done by yourself under the **supervision of Mr. Md. Eftekhar Alam** Dept. Of EEE, IIUC and **Co Supervision of Prof. Mohammed Shamsul Alam** Dept. Of CSE, IIUC. No portion of the work contained in this Project has been submitted in support of any application for any other degree or qualification of this or any other university or Institute of learning.

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ABSTRACT

We are stuck with technology when what we really want is just stuff that works. With the current paradigm shift in technological field, there is an urgent need to embrace and appreciate the power of technology. Housing sector remains vigilant to face the challenges of change by employing a new strategy that facilitates easy management of apartments. Hence there is need to develop a “**Apartment Management System**” that can simplify work for the rental admins so that all their work can be efficient and effective. To get information about how apartments are currently being managed, I prepared questionnaires and submitted them to a number of apartment admins and from the information I gathered I realized all work was done manually with a lot of paper work involved. Papers can easily get damaged or get lost leading to loss of data. It is also expensive to keep on buying files to store your records. A lot of files make a place look untidy and also consume a lot of space. Getting a certain file to check data from many files becomes a difficult task. Considering those facts, I decided to develop a apartment management system that can solve all the problems experienced with the current manual system. The system was developed in such manner that it provides maximum user-friendly interface using **PHP** [Hypertext pre-processor] programming language with **Laravel** using the **XAMPP** [Cross-Platform (X), Apache (A), MariaDB (M), PHP (P) and Perl (P)] as the testbed.

CHAPTER-1

1. INTRODUCTION

1.1 Introduction:

This is such a system which can provide apartment rent facility to the single family, Large family & bachelors. The existing systems such as “online rent system” which can provide the facilities from any place with very low cost. This system is very much efficient and effective because it is fully automated. Our main aim to build a complete “**Apartment Management System**” successfully which will be fully automated online rent system with very low cost. This system will ensure that they can get their apartments without going to the building physically.

1.2 Motivation:

In the case of reality of Bangladesh rental system, everybody searches the rent manually but most of educated person are used to internet. So, we believe that there is a demand for an online system that answers customers query related finding flat and other type of accommodation. Information and data searching online has become increasingly popular over the last few years. Everyone used to feel comfort to collect information by using online. But they do not get the user-friendly rental system for rent of all kinds of accommodation. So, we made “Apartment Management System” where every person can get all kinds of accommodation information for rent using internet browsing of our site.

1.3 Existing System:

Currently, the maximum homes admins manipulate residences and renter’s details on papers. With the present-day device recording the information of diverse activities of a person is completely manual and a whole lot of paper paintings. Each house has a record that includes the residence: number, length, rent consistent with month, centers, and status. Rent price desk contains renters: first name, remaining name, Phone wide variety, date of fee, amount, and stability if any.

The existing system most effective gives a text-based interface that is not as user pleasant as a Graphical consumer interface. Since the system is applied manually, the response may be very sluggish. The transactions aren't at ease as papers may wander away or damaged. Hence, there is a need for reformation of the gadget with greater benefits and versatility. The device removes most of the restrictions of the present machine.

1.4 Our system:

The main goal of the system is to create automated process carried out in the system with improved performance and realize the vision of paperless work. There are different modules such as owner, admin and renter. They played separate area in the system. Owner can view all work of the system, take action and give access to admin for manage other sectors. Admin can view the renter site and manage all activities of renter & watchman. The system which can design and develop functional management and service maintenance. We can measure apartment details, location control, monthly maintenance expenses or bill, parking, events manage, handling notice, Unknown entry selfie (for extra security), swimming system(optional), complain, some important links. Renter can view all his features and book events, swimming system, parking allocation, give complain and reviews. The report of billing is search by date and also see the list of total transactions. Multi building task are handling with a centralized database management system.

1.5 Objectives:

The system will be password-protected. AMS will be a multi-user system where every user must log in. AMS need to perform the following objectives:

- ☐ To develop an apartment management system that allows the user to view renter's data as well as building record.
- ☐ To develop a system that allows the users to add, edit, search and delete data from the database.
- ☐ To study and analysis, the requirement specifications of the apartment management system.

1.6 Summary:

We have described the whole process such as the existing system, our system, objectives in this chapter. Also, the different module in our system and their role and worker described.

CHAPTER-2

2. LITERATURE REVIEW OF PROJECT

2.1 Introduction:

This chapter mentions about literature review of our system, the definition of literature review, similar system and comparison of our system.

2.2 Literature Review:

A literature evaluation is a powerful evaluation of selected documents on a research topic. An evaluation may additionally form a vital part of the study's manner or may represent a challenge in itself. In the following literature review, we are able to discuss a few present structures that are partially similar to our mission.

2.3 Similar System:

BDto-let.com is web-based software and is designed to help search houses for sale and rent. This website mainly used for developers for his/her own construction buildings for sale and rent. . Here developer corporation at once involved in rent and sell. This website only for using house rent but user want a site which provides all rental services.

2.4 Comparisons with our System:

In this section we will make a comparison of the following system with our system. In the previous section, we saw that the system has issues with various content such as recommendation system, system specification etc. In our system, we have made a proper system for Apartment. It also user-friendly and there are many features and works. We use Xampp database to get the proper advantages of server-side programming. In our system, we are trying to give proper administration based on the renter facilities. We are trying to give almost all facilities based on the renter needs. There is also event system (for small functions), swimming system(optional), unknown entry selfie (for security), Notice & Complain system. This all our new focused area of this system. We also add the E-Mail notification and SMS process for the Renter reminders.

2.5 Summary:

In this chapter, we describe the literature review and the similar application of our system. We also describe comparisons with our system.

CHAPTER-3

3. METHODOLOGY

3.1 Introduction of Methodology:

The term technique method the approach and procedure followed through undertaking a study observation. It outlines how statistics might be amassed and the tools for collecting records, machine methodology, the proposed system input and output, users, and systems development equipment.

3.2 Facts finding techniques:

It indicates how records may be accumulated from the customers of the gadget. The records collection techniques for use include:

- ☐ It will use this approach to accumulate facts about how the cutting-edge gadget operates and its tactics. This entails systematically watching and recording the behavior and traits of operations and techniques.
- ☐ It gives extra distinct and context-related information and might adapt to events as they arise however the technique may be time consuming.

3.3 Software Development Life Cycle (SDLC):

Software Development Life Cycle (SDLC) offers us an overview and hints to develop satisfactory software. Whenever SDLC is hired, the aim is to create a gadget that meets the number one proprietor's expectancies. For the "Apartment Management System", we comply with the Software Development Life Cycle to make our internet site dependable for the user.

3.3.1 Software process model:

Process is an activity, which operates on an object and changes its nation. Model is the graphical illustration of an item. So, a technique model shows the sports of software program graphically. We need process model because manner is critical than product (software). If the system is ideal, the product also is right.

3.3.2 Feasibility Study:

Feasibility observe is the excessive-degree tablet version of the complete requirement analysis technique. The goal of the feasibility look is to determine whether the proposed device can be advanced with available resources.

There are 4 steps to be followed for determining the feasibility study of the proposed system.

- ☐ Technical Feasibility
- ☐ Operational Feasibility
- ☐ Economic Feasibility
- ☐ Social Feasibility

3.3.3 Technical Feasibility:

- ☐ It centers on the prevailing computer system. (Hardware/Software) and to what quantity it is able to guide the proposed addition.
- ☐ The facility to provide outputs in a given time.
- ☐ Response time beneath sure situations.
- ☐ Ability to procedure a positive volume of transactions at a particular pace.
- ☐ Facility to communicate records to a distant place.

3.3.4 Operational Feasibility:

- ☐ What modifications may be bought with the gadget?
- ☐ What organizational systems are disbursed?
- ☐ What new competencies will be required?

3.3.5 Economic Feasibility:

- ☐ Economic analysis is the maximum regularly used method for evaluating the effectiveness of a brand-new gadget.
- ☐ More usually called price/benefit evaluation. The method is to determine the blessings and savings which might be expected from a proposed system and examine them with expenses.
- ☐ The improvement cost for the improvement of the proposed system is true.

3.3.6 Social Feasibility:

- ☐ What modifications may be offered with the system?
- ☐ What organizational structures are distributed?
- ☐ What new abilities can be required?

3.4 Requirement Definition:

At this step, we have to develop system requirements by using well-known requirement engineering tests. Requirement engineering has three parts that are:

- ☐ Requirement Analysis
- ☐ Requirement Specification
- ☐ Requirement Validation

3.5System Specification:

At this segment, we focus on ER-Diagram which shows the entity, attribute, and relationship with involving one to one, one to many, many to many, assign primary keys, foreign keys and perform normalization to reduce redundancy.

3.1 Implementation:

Numerous systems are made for Apartment Management System yet we attempted to an added couple of new highlights to make adaptable utilization of the clients. We utilized the social database to solid, recovering information for client and looking wind up dependable and less demanding. In execution of the framework plan and coding stage are vital.

3.2Summary:

In this chapter, we describe the process model of our system. We use this Agile model because it is easily understanding about the requirement needed for the system. We must follow SDLC to develop our system.

CHAPTER-4

4. DATABASE DESIGN

4.1 Database Design:

A database is a collection of facts this is organized in order that it could be without problems accessed, managed, and up to date. Data is organized into rows, columns, and tables, and it is listed to make it less difficult to locate applicable statistics. Data receive up to date, accelerated, and deleted as new records are brought. Database system workloads to create and updates themselves, querying the records they comprise and jogging packages towards it. Computer databases generally include aggregations of statistics facts or documents, inclusive of sales, transactions, product catalogs and inventories, and purchaser profiles. Typically, a database supervisor gives customers the capacity to manipulate read/write get entry to, specify record technology, and examine utilization.

4.2 Entity Relationship Diagram:

Entity relationship model is used to represent the conceptual schema of the database. The important method of entity relationship model is entity relationship model in which set of entities are represented by relation in a graphical form .In an effective system, data is divides into discrete categories or entities .An entity relationship (E-R) model is an illustration of various entities in a business and the relationship between them and ER model is derived from business specification or narratives and build during the analysis phase of the system development life cycle.ER model separates the information required by business from the activities performed within a business.

Symbols And Meaning Of ERD:

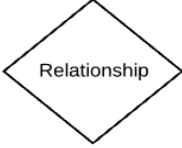
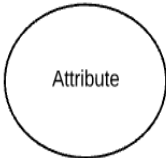
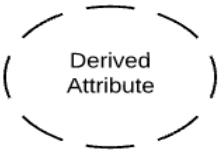
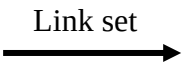
Relationship Symbols	Meaning
	Relationships are associations between or among entities.
	Weak Relationships are connections between a weak entity and its owner.
	Attributes are characteristics of an entity, a many-to-many relationship, or a one-to-one relationship.
	Multivalued attributes are those that are can take on more than one value.
	Derived attributes are attributing whose value can be calculated from related attribute values.
	For making connection.

Table 4-1 SYMBOLS AND MEANING OF ERD

4.2.1 AMS ER-Diagram:

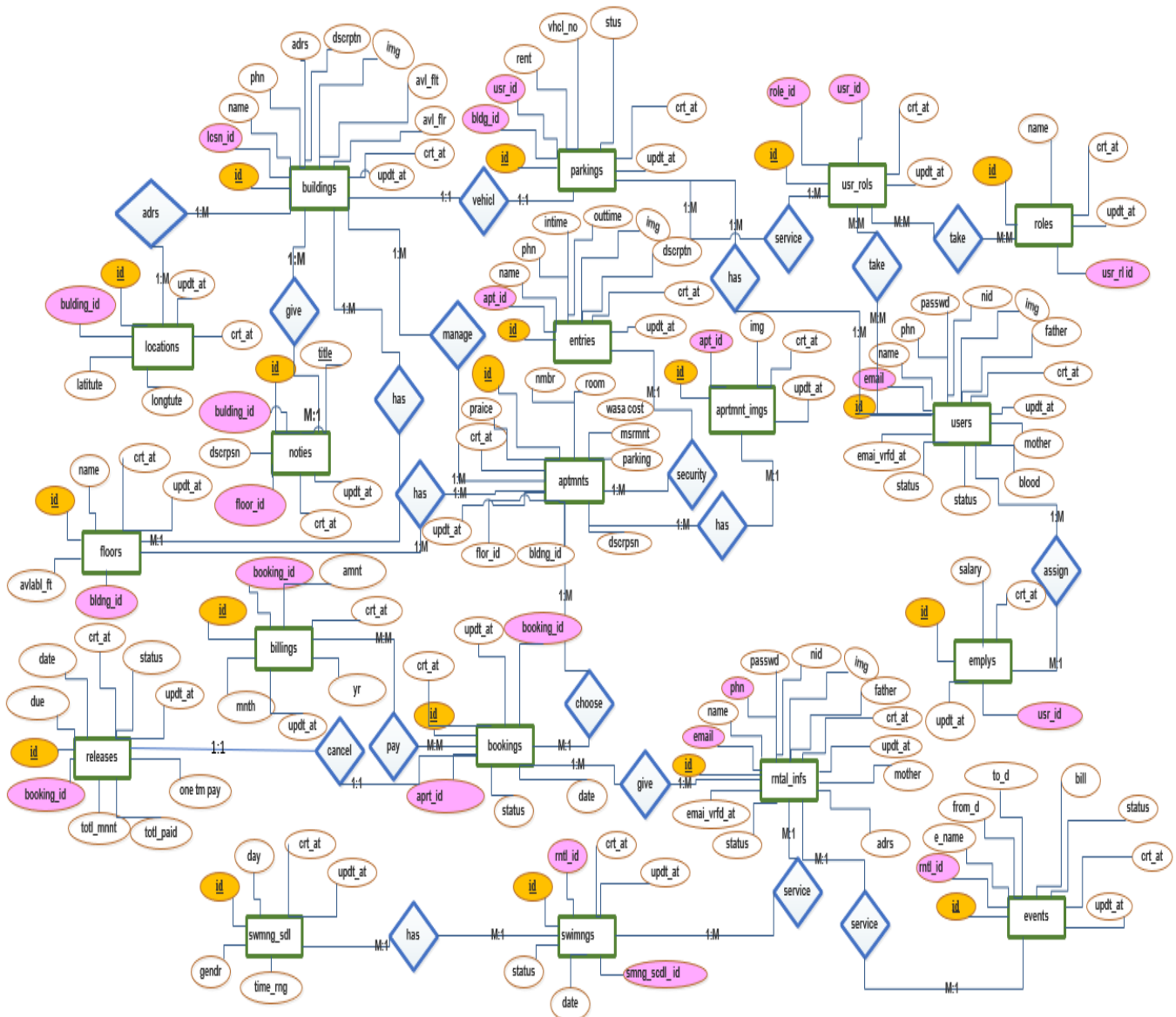


Figure 4.2.1: ERD Diagram

4.3 Summarization of ERD:

From apartment Management system's ERD, we can identify the entities, attributes and relationships. We can see that the entities are billings, Apartments, user, cities, wards, thanas, swimming, parking, event, booking, renters, billing.

In relationship table relationship names are:

Table 4-2 Summarization of ERD

Relationship Name	Relationship Type
Role to Owner	One to one
Role to Admin	One to one
Role to Renter	One to many
Building to location	One to many
Building to cities	Many to one
Building to wards	Many to one
Building to thanas	Many to one
Building to user	Many to many
Building to apartment	Many to one
Swimming service for users	Many to one
event service for users	Many to one
Parking service for users	Many to one
Renter involve in users	Many to many
Billing for booking	Many to many
Cities to buildings	One to many
Ward to building	One to many
Thanas to building	One to many
Users to buildings	Many to many
Apartment to building	One to many
Users for swimming service	One to many
Users for events service	One to many
User for parking service	One to many
Booking for billing	Many to many

Table 4-2: Summarization of ERD

Table 4-3 Apartment Management System:

Table 4-3 TBL Building:

Field Name	Data Type	Value	Key
id	int(10)	Not Null	Primary Key
name	varchar(191)	Not Null	
Phone	text	Not Null	
Address	text	Null	
description	text	Null	
Image	varchar(191)	Null	
available_flat	varchar(191)	Not Null	
available_floor	varchar(191)	Not Null	
city_id	int(10)	Not Null	Foreign Key
thana_id	int(10)	Not Null	Foreign Key
ward_id	int(10)	Not Null	Foreign Key
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-3: Building Table

Table 4-4 TBL Apartment:

Field Name	Data Type	Value	Key
id	int(10)	Not Null	Primary key
number	varchar(191)	Not Null	
type	varchar(191)	Not Null	
Room	varchar(191)	Not Null	
bed_room	varchar(191)	Not Null	
bath_room	varchar(191)	Not Null	
price	double(8,2)	Not Null	
wasa_cost	int(11)	Not Null	
dinning_room	varchar(191)	Null	
drawing_room	varchar(191)	Null	
balcony	varchar(191)	Null	
measurement	varchar(190)	Null	
parking	varchar(191)	Null	
description	text	Null	
available	varchar(191)	Null	
building_id	int(10)	Not Null	Foreign Key
floor_id	int(10)	Not Null	Foreign key
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-4: Apartment Table

Table 4-5 TBL Apartment Images:

Field Name	Data Type	Value	Key
id	int(10)	Not Null	
image	varchar(191)	Not Null	
apartment_id	int(10)	Not Null	Foreign key
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4.-5: Apartment Images Table

Table 4-6 TBL Bookings:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary Key
renter_id	int(10)	Not Null	Foreign key
apartment_id	int(10)	Not Null	Foreign Key
date	varchar(191)	Null	
status	int(1)	Not Null	
advance	double(8,2)	Null	
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-6: Bookings Table

Table 4-7 TBL Bookings:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary Key
booking_id	int(10)	Not Null	Foreign Key
month	varchar(191)	Not Null	
year	varchar(191)	Not Null	
amount	double(8,2)	Not Null	
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-7: Billing Table

Table 4-8 TBL Cities:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary Key
name	varchar(191)	Not Null	
latitude	double	Null	
longitude	double	Null	
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-8: Cities Table

Table 4-9 TBL Wards:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary Key
name	varchar(191)	Not Null	
latitude	double	Null	
Longitude	double	Null	
thana_id	int(10)	Not Null	Foreign Key
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-9: Wards Table

Table 4-10 TBL Thanas:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary Key
name	varchar(191)	Not Null	
latitude	double	Null	
longitude	double	Null	
city_id	int(10)	Not Null	Foreign Key
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-10: Thana's Table

Table 4-11 TBL Renters:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary Key
name	varchar(191)	Not Null	
email	varchar(191)	Not Null	Foreign Key
password	varchar(191)	Not Null	
nid	varchar(191)	Not Null	
father	varchar(191)	Not Null	
mother	varchar(191)	Not Null	
mobile	varchar(191)	Not Null	Foreign Key
address	varchar(191)	Null	
image	varchar(191)	Null	
status	tinyint(1)	Null 0	
email_verified_at	timestamp	Null	
remember_token	varchar(100)	Null	
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-11: Renters Table

Table 4-12 TBL Users:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
Name	varchar(191)	Not Null	
email	varchar(191)	Not Null	Foreign Key
password	varchar(191)	Not Null	
nid	varchar(191)	Not Null	
father	varchar(191)	Not Null	
mother	varchar(191)	Not Null	
mobile	varchar(191)	Not Null	
address	varchar(191)	Null	
Image	varchar(191)	Null	
Blood	varchar(10)	Null	
Status	int(11)	Null	
email_verified_at	timestamp	Null	
remember_token	varchar(100)	Null	
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-12: Users Table

Table 4-13 TBL Swimming:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
renter_id	int(10)	Not Null	Foreign Key
swimming_schedule_id	int(10)	Not Null	Foreign Key
Date	date	Not Null	
status	int(11)	Null	
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-13: Swimming Table**Table 4-14 TBL Parking:**

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
building_id	int(10)	Not Null	Foreign Key
vehicle_no	varchar(191)	Not Null	
Rent	double(8,2)	Not Null	
user_id	int(10)	Not Null	
status	tinyint(1)	Not Null	
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-14: Parking Table

Table 4-15 TBL Events:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
renter_id	int(10)	Not Null	Foreign Key
event_name	varchar(191)	Not Null	
food_menu	varchar(191)	Not Null	
from_date	date	Not Null	
to_date	date	Not Null	
bill	int(11)	Null	
status	int(11)	Null	
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-15: Events Table**Table 4-16 TBL Companies:**

Field Name	Data Type	Value	Type
id	int(10)	Null	Primary key
about	longtext	Null	
phone	varchar(191)	Null	
phone1	varchar(191)	Null	
mail	varchar(191)	Null	
address	varchar(191)	Null	
facebook	varchar(191)	Null	
twitter	varchar(191)	Null	
instagram	varchar(191)	Null	
linkedin	varchar(191)	Null	
gmail	varchar(191)	Null	
whatsapp	varchar(191)	Null	
logo	varchar(191)	Null	

favicon	varchar(191)	Null	
mobile_logo	varchar(191)	Null	
footer_logo	varchar(191)	Null	
name	varchar(191)	Null	
download	longtext	Null	
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-16:Companies Table

Table 4-17 TBL Employees:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
user_id	int(10)	Not Null	Foreign Key
salary	int(11)	Not Null	
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-17: Employees Table

Table 4-18 TBL Entries:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
name	varchar(191)	Not NULL	
mobile	varchar(191)	Null	
apartment_id	int(10)	Not Null	Foreign Key
image	varchar(191)	Not NULL	

intime	datetime	NULL	
outtime	datetime	NULL	
description	varchar(191)	NULL	
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

Table 4-18: Entries Table

Table 4-19 TBL Floors:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
name	varchar(191)	Not Null	
available_flat	varchar(191)	Not Null	
building_id	int(10)	Not Null	Foreign Key
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

Table 4-19: Floors Table

Table 4-20 TBL Linkups:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
name	varchar(191)	Not Null	
mobile	varchar(191)	Not Null	
address	varchar(191)	Not Null	
link	varchar(191)	Not Null	
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

Table 4-20: Linkups Table

Table 4-21 TBL Messages:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
name	varchar(191)	Not Null	
phone	varchar(191)	Not Null	
email	varchar(191)	Not Null	
details	varchar(5000)	Not Null	
created_at	timestamp	NULL	
updated_at	Timestamp	NULL	

Table 4-21: Messages Table**Table 4-22 TBL Migrations:**

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
migration	varchar(191)	Not Null	
batch	int(11)	Not Null	

Table 4-22: Migrations Table**Table 4-23 TBL News links:**

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
heading	varchar(191)	Not Null	
link	Longtext	Not Null	
image	varchar(191)	Not Null	
date	Date	Not Null	
created_at	Timestamp	NULL	
updated_at	Timestamp	NULL	

Table 4-23: News links Table

Table 4-24 TBL Notices

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
building_id	int(10)	Not Null	Foreign Key
floor_id	int(10)	NULL	Foreign Key
title	varchar(191)	NULL	
description	varchar(191)	Not Null	
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

Table 4-24: Notices Table**Table 4-25 Password Resets:**

Field Name	Data Type	Value	Type
email	varchar(191)	Not Null	Foreign key
token	varchar(191)	Not Null	
created_at	timestamp	NULL	

Table 4-25: Password Resets Table**Table 4-26 TBL Payments Reports:**

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
billing_id	int(10)	Not Null	
amount	double(8,2)	Not Null	
due	double(8,2)	Not Null	
date	date	Not Null	
created_at	timestamp	Null	
updated_at	timestamp	Null	

Table 4-26: Payments Reports Table

Table 4-27 TBL Releases:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
booking_id	int(10)	Not Null	Foreign key
total_amount	double(8,2)	Not Null	
total_paid	double(8,2)	Not Null	
onetime_pay	double(8,2)	NULL	
due	double(8,2)	NULL	
date	date	NULL	
status	tinyint(1)	NULL	
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

4-27 :Releases Table**Table 4-28 TBL Reviews:**

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
user_id	int(10)	Not Null	
apartment_id	int(10)	Not Null	
display_name	varchar(191)	NULL	
review	varchar(191)	NULL	
point	int(11)	None	
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

4-28 :Reviews Table

Table 4-29 TBL Roles:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
user_id	int(10)	Not Null	
apartment_id	int(10)	Not Null	
display_name	varchar(191)	NULL	
review	varchar(191)	NULL	
point	int(11)	None	
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

4-29 :Roles Table**Table 4-30 TBL Sliders:**

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
title	varchar(191)	Not Null	
description	varchar(191)	Not Null	
image	varchar(191)	Not Null	
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

4-30 :Sliders Table

Table 4-31 TBL Swimming Schedule:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
gender	varchar(191)	Not Null	
day	varchar(191)	Not Null	
time_range	varchar(191)	Not Null	
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

4-31: Swimming Schedule Table**Table 4-32 TBL Tips:**

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
title	varchar(191)	Not Null	
description	varchar(191)	Not Null	
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

4-32 :Tips Table

Table 4-33 TBL User Roles:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
user_id	int(10)	Not Null	Foreign key
role_id	int(10)	Not Null	Foreign key
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

4-33: User Roles Table

Table 4-34 TBL Work Schedules:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
user_id	int(10)	Not Null	Foreign key
shift	varchar(191)	NULL	
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

4-34: Work Schedules Table

Table 4-35 TBL Complains:

Field Name	Data Type	Value	Type
id	int(10)	Not Null	Primary key
user_id	int(10)	Not Null	Foreign key

details	text	Not Null	
created_at	timestamp	NULL	
updated_at	timestamp	NULL	

4-35: TBL Complains Table

4.5 Summary:

In this chapter we discussed about data dictionary, database management system and tables of Apartment Management System. We have written symbol and meaning of database. We are design ERD and short description.

CHAPTER-5

5. SOFTWARE DESIGN

5.1 Introduction:

Software Design is a process by means of which the Software necessities are translated into a representation of software program components interfaces and records necessary for the implementation section. The software layout description describes the design of the difficulty of hobbies of a laptop software configuration item (CSCI). It describes the CSCI -huge layout choices, the CSCI architectural design, and the distinctive design had to enforce the software program. An SDD is a report used to specify gadget architecture and application design in a software program related challenge. The reason for the software design description is to file statistics that:

- ☐ Describe the useful shape, records, and set of rules to be applied.
- ☐ Identify the required device resources.
- ☐ Can be used to assess the effect of requirement changes.
- ☐ Assist in generating take a look at instances.
- ☐ Can be used to confirm compliance with necessities.

5.2 Overall design:

5.2.1 Use Case Diagram:

A Use case is simple method which is used to identify and organize the system requirements in system analysis. This method is made up of the possible sequences of interactions between the systems and users in a particular goal.

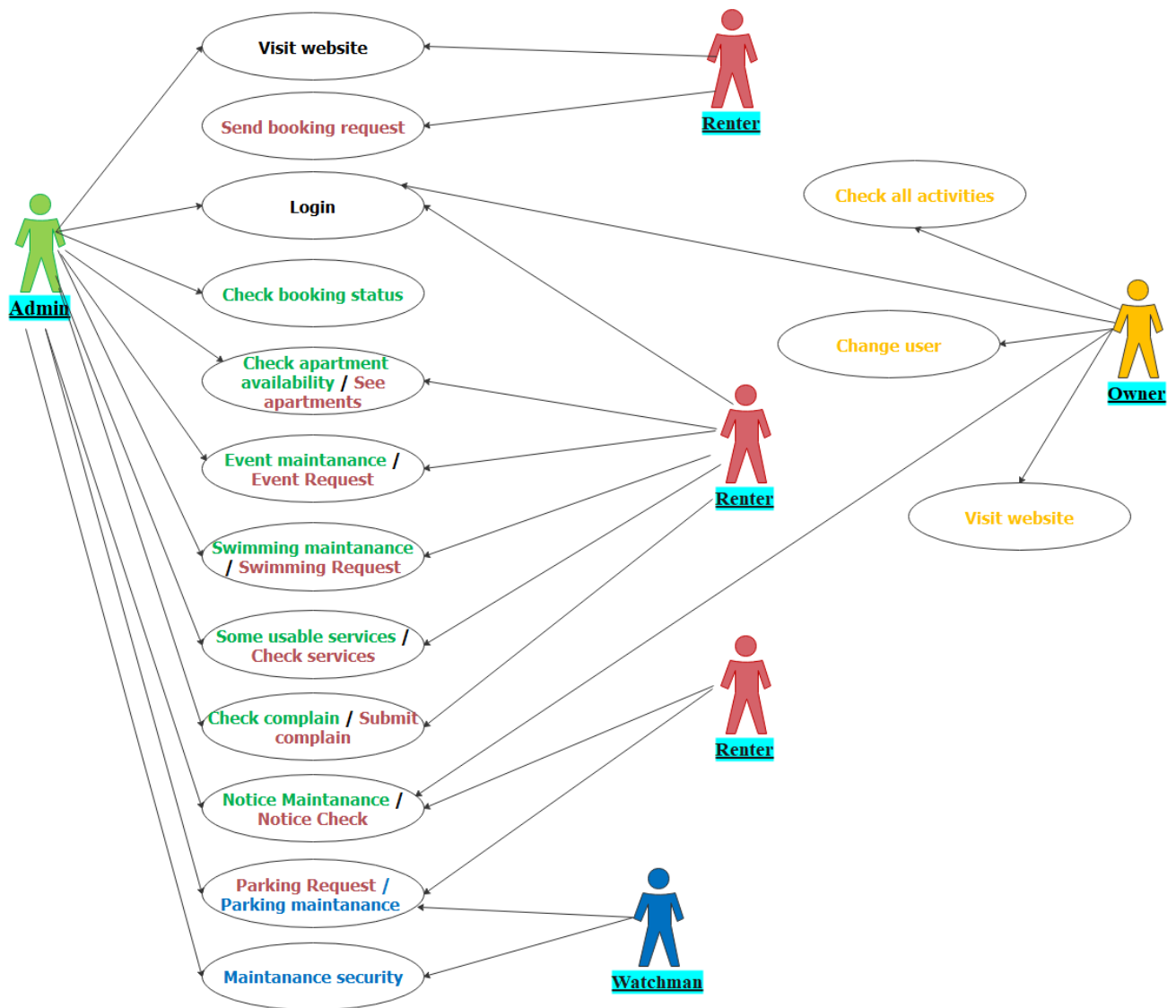


Figure 5.2.1: Use Case Diagram

5.2.2 Activity Diagram:

5.2.2.1 Owner Activity diagram:

If Username and password is correct owner can login to the system otherwise shows error login and refresh page. The system then checks the access level of owner if the owner login will be successful and he will redirect to the dashboard as well as access level. Then owner can manage user, building, billing, notice, services complain.

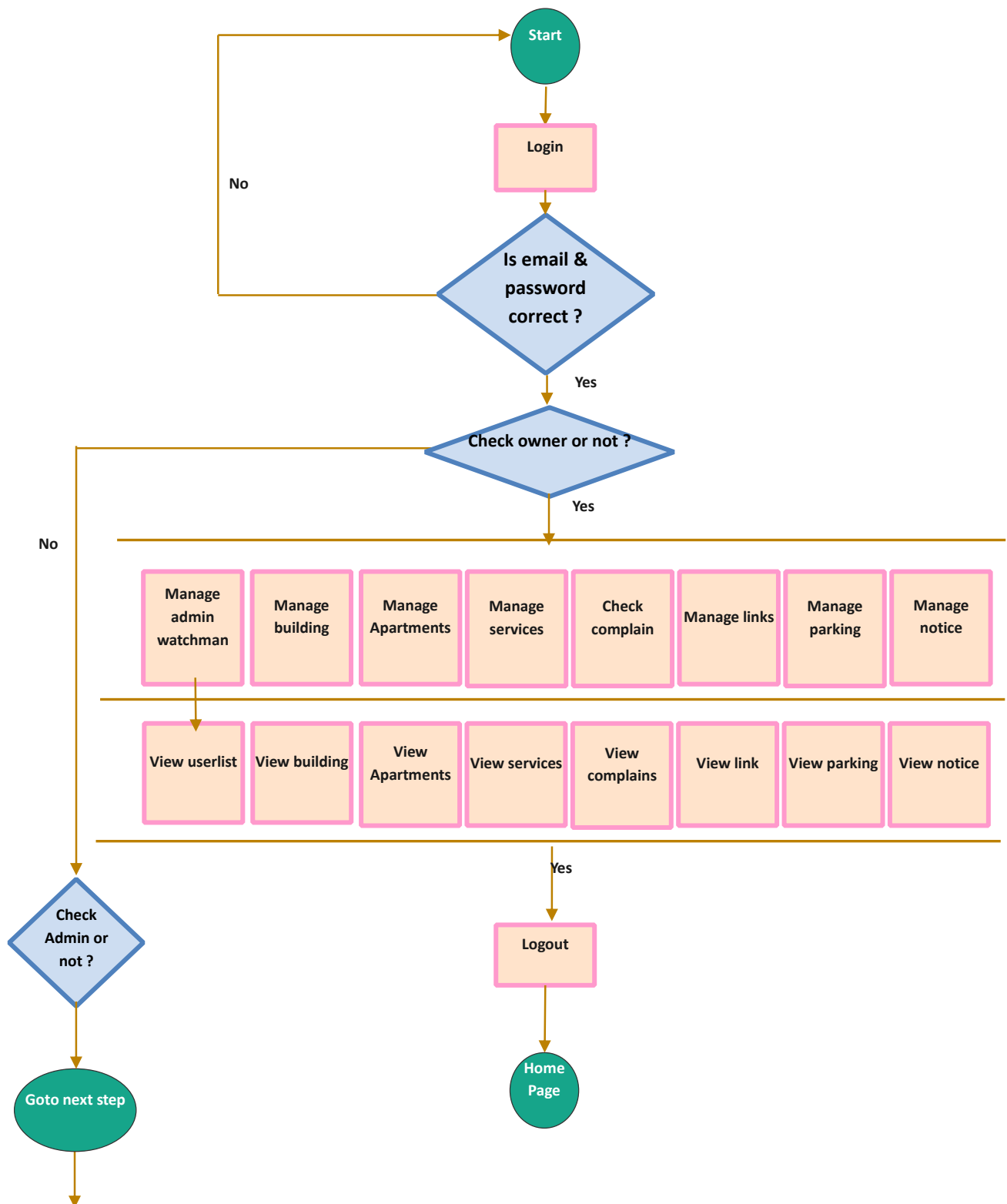


Figure 5.2.2.1: Owner Activity Diagram

5.2.2.2 Admin Activity Diagram:

If Username and password is correct admin can login to the system otherwise shows error login and refresh page. The system then checks the access level of admin if the admin login will be successful and he will redirect to the dashboard as well as access level. Then admin can manage renter, building, billing, Apartment complain, notice, events, watchmen, swimming system reports.

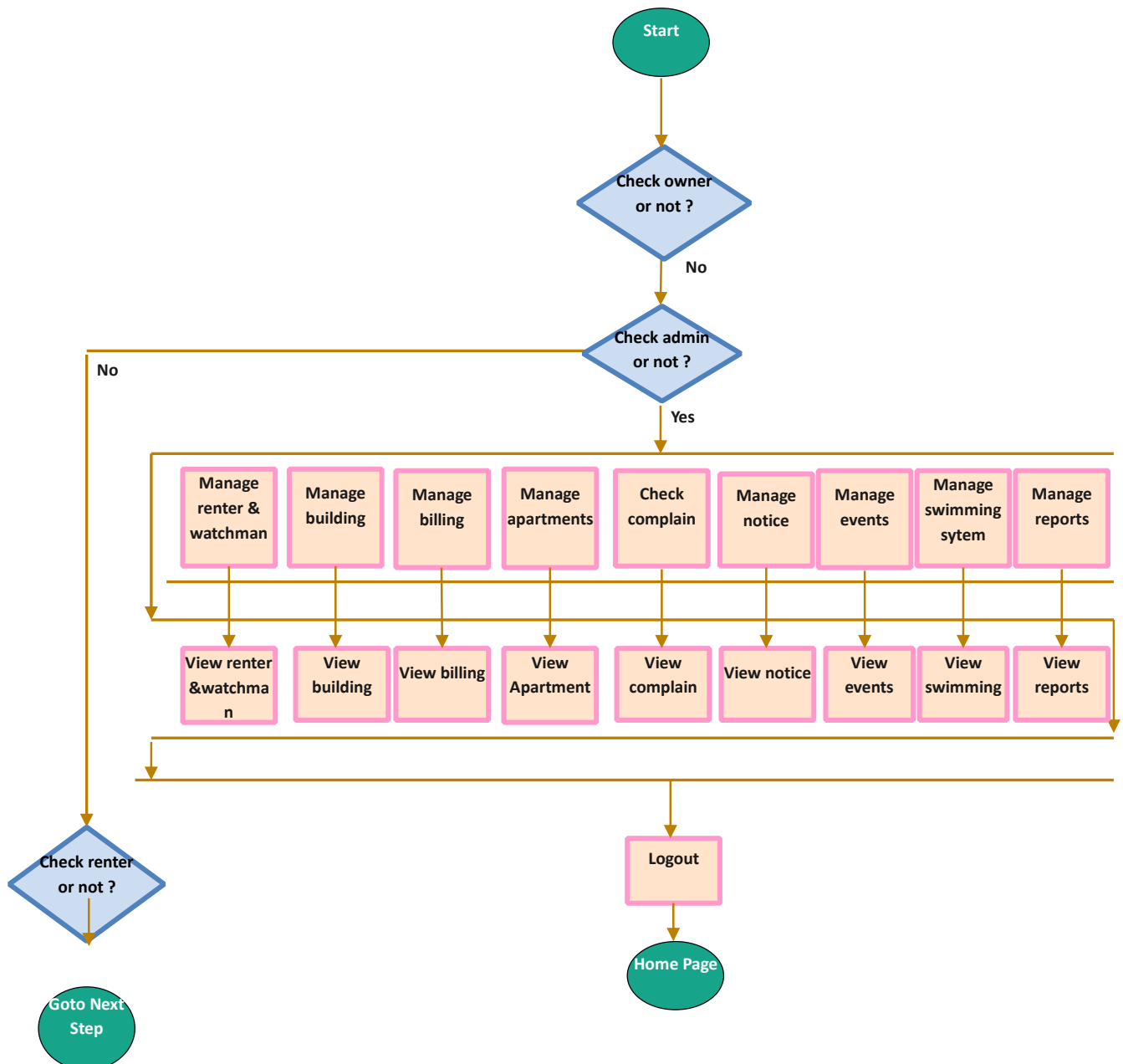


Figure 5.2.2.2: Admin Activity Diagram

5.2.2.3 Renter Activity Diagram:

If Username and password is correct renter can login to the system otherwise shows error login and refresh page. The system then checks the access level of renter if the renter login will be successful and he will redirect to the dashboard as well as access level. Renter can see billing information, services information, choose billing report, finding notice, see apartment details, manage complain.

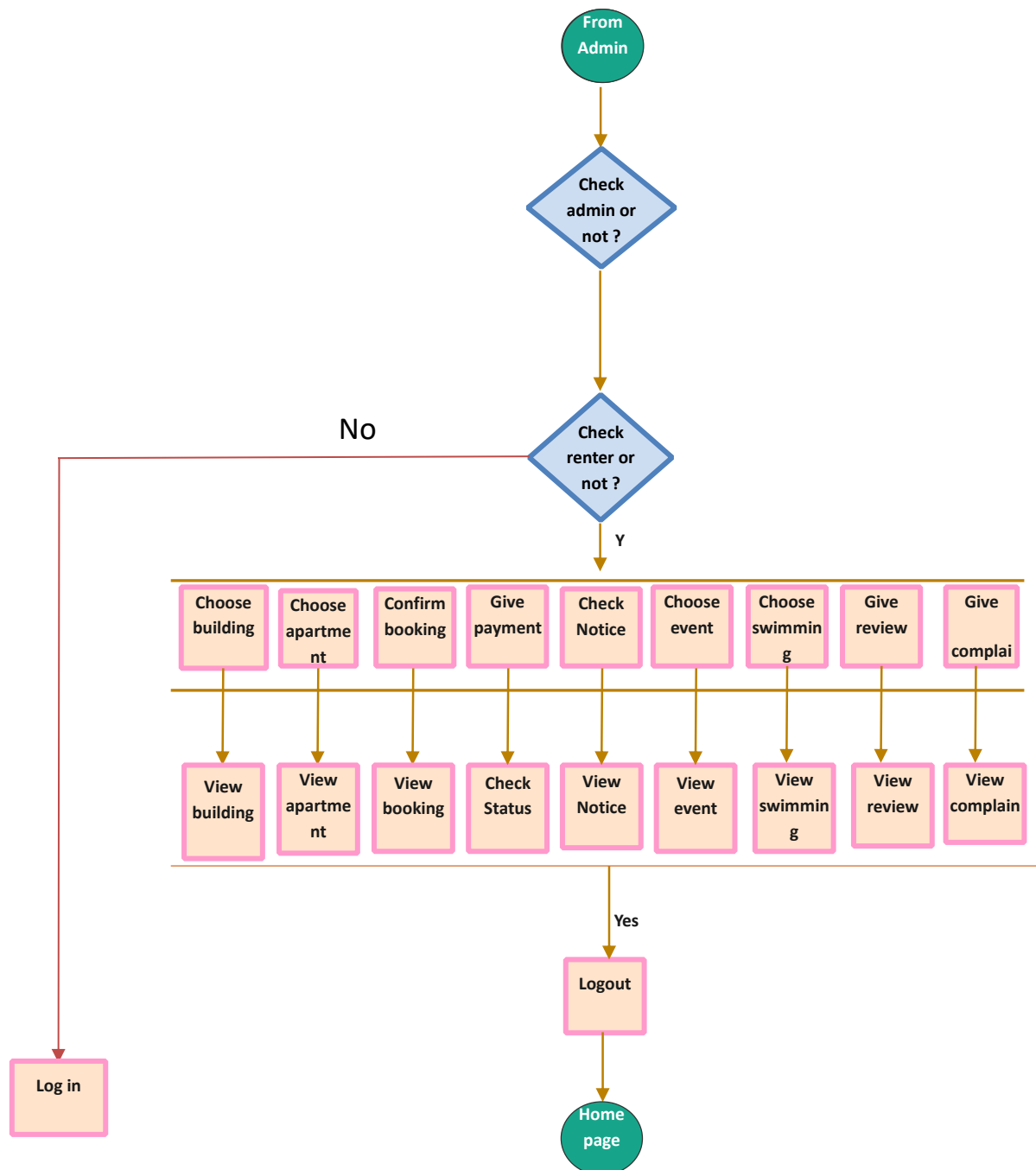


Figure 5.2.2.3: Renter Activity Diagram

5.2.3 Data Flow Diagram (DFD):

Data flow diagrams are quite important for the requirement specification of the device. Data flow diagrams are an intuitive manner of displaying how records are processed via a system. Those are used to reveal how data flows through a series of processing steps. The records are converted at every step or differences are application features while statistics flow diagrams are used to document a system.

5.2.3.1 Context Diagram:

A context diagram is an image design that clarifies the interfaces and obstacles of the venture or procedure to hand. It no longer handiest suggests the technique or venture in its context, it additionally suggests the venture's interactions with different systems and customers. A System Context Diagram (SCD) in software engineering and systems engineering is a diagram that represents the actors outdoor a gadget that might have interaction with that machine. This diagram is the very best-stage view of a device. It is similar to a Block diagram.

A. Level 0:

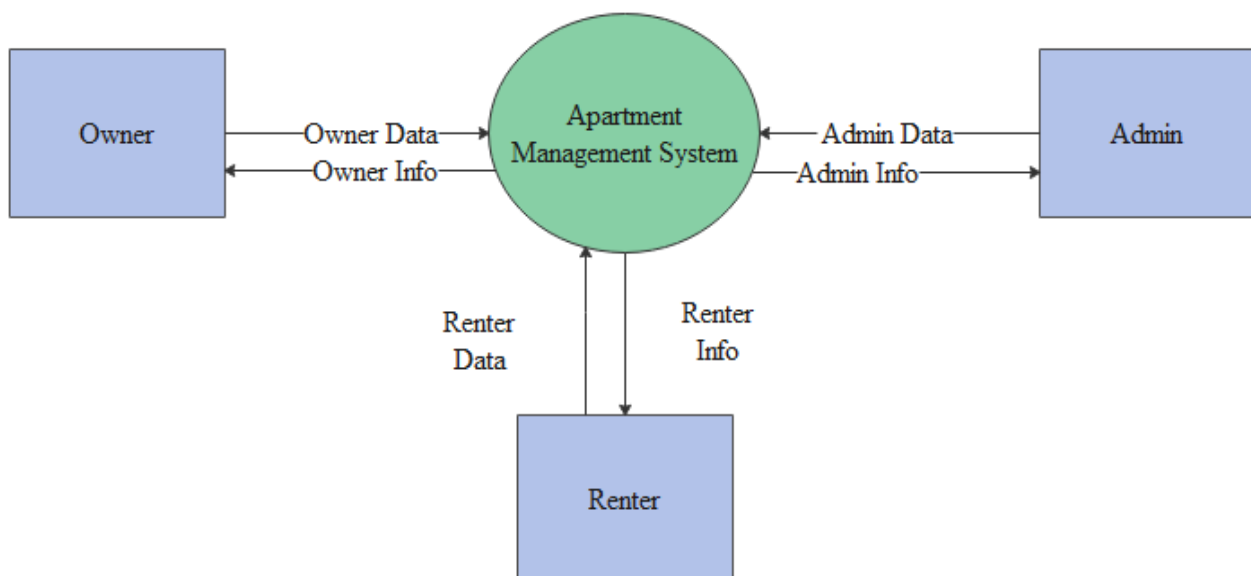


Figure 5.2.2.4: Level 0

B. Level 1(Owner):

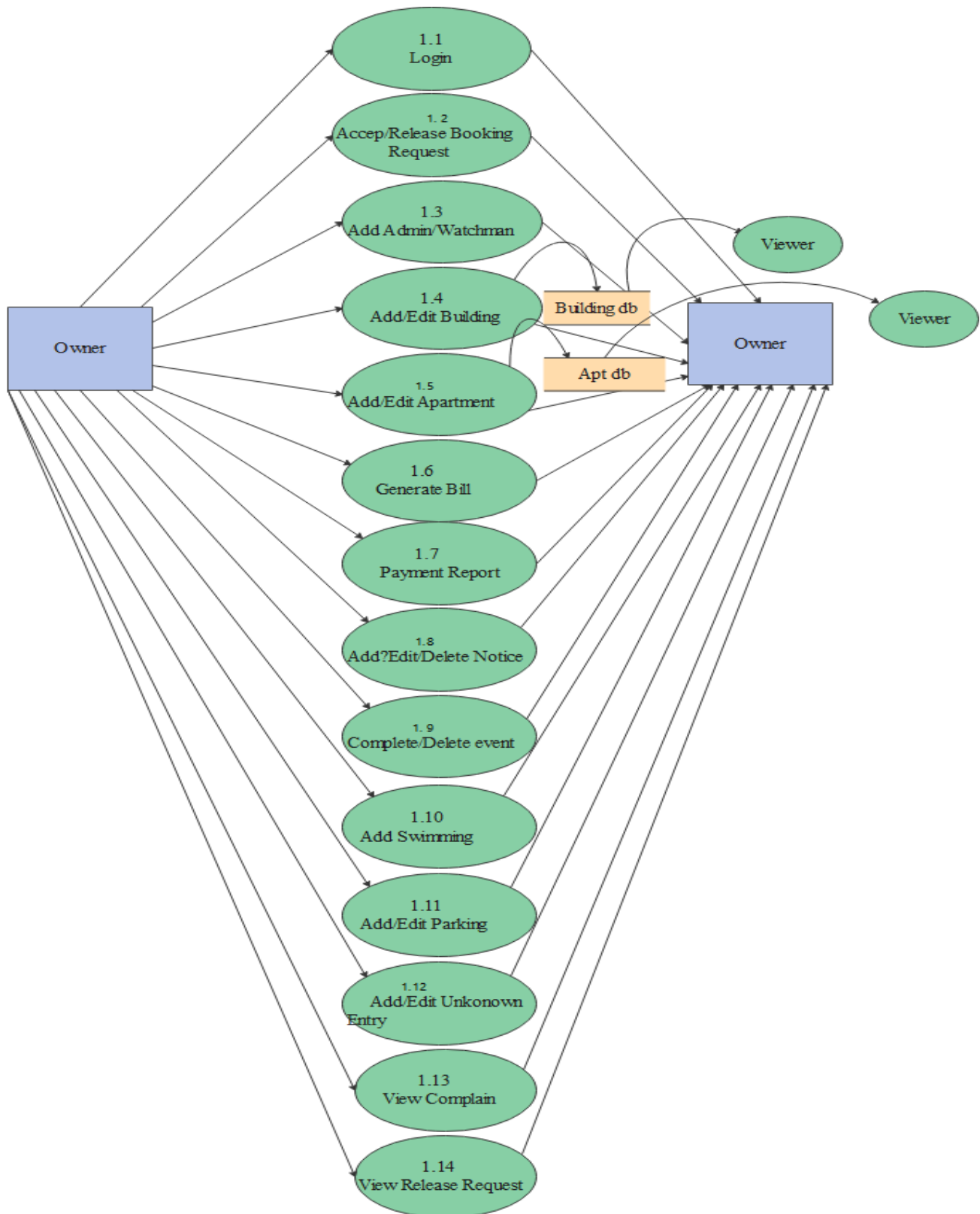


Figure 5.2.2.5: Level 1(Owner)

C. Level 1(Admin):

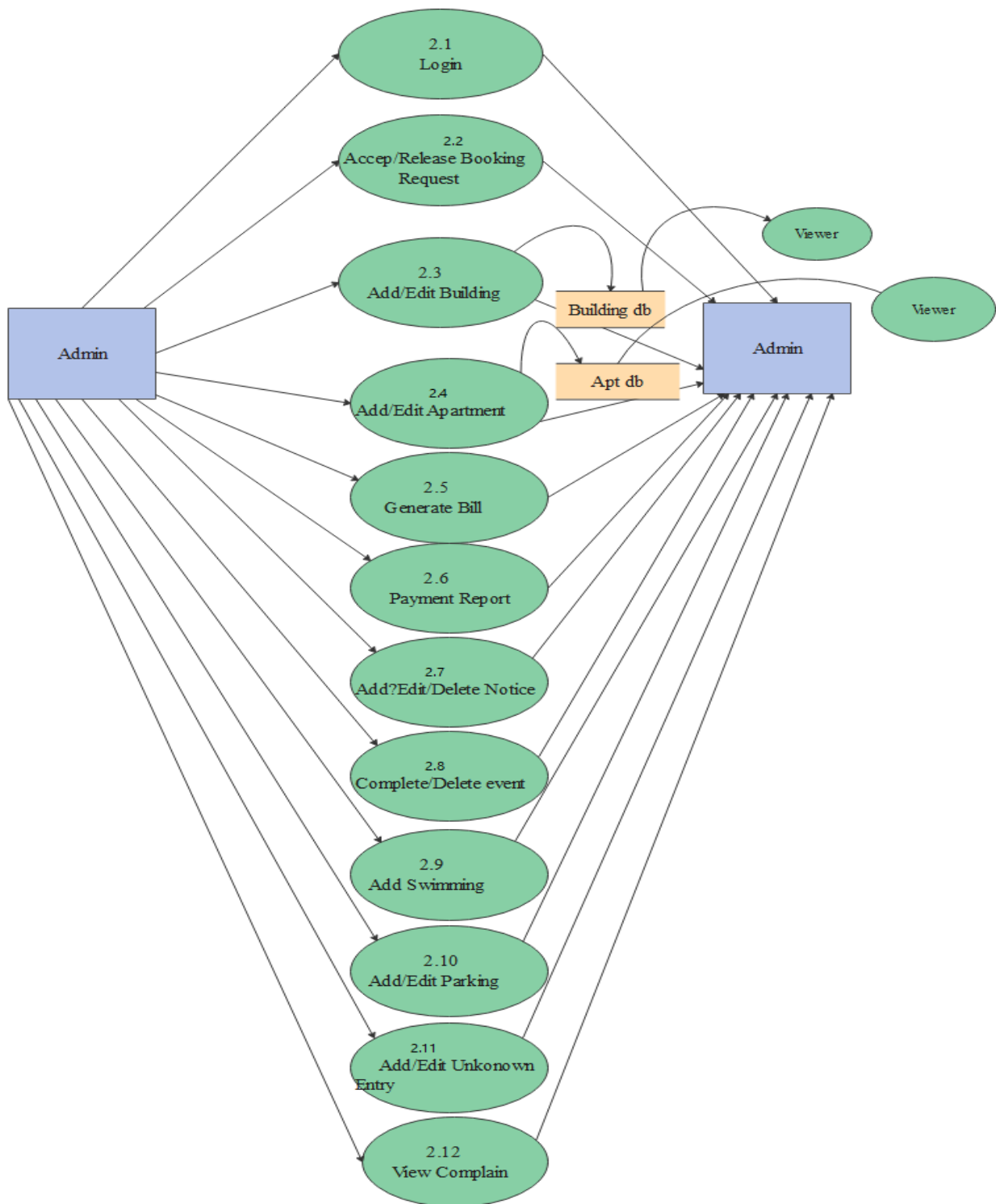


Figure 5.2.2.6: Level 1 (Admin)

D. Level 1(Renter):

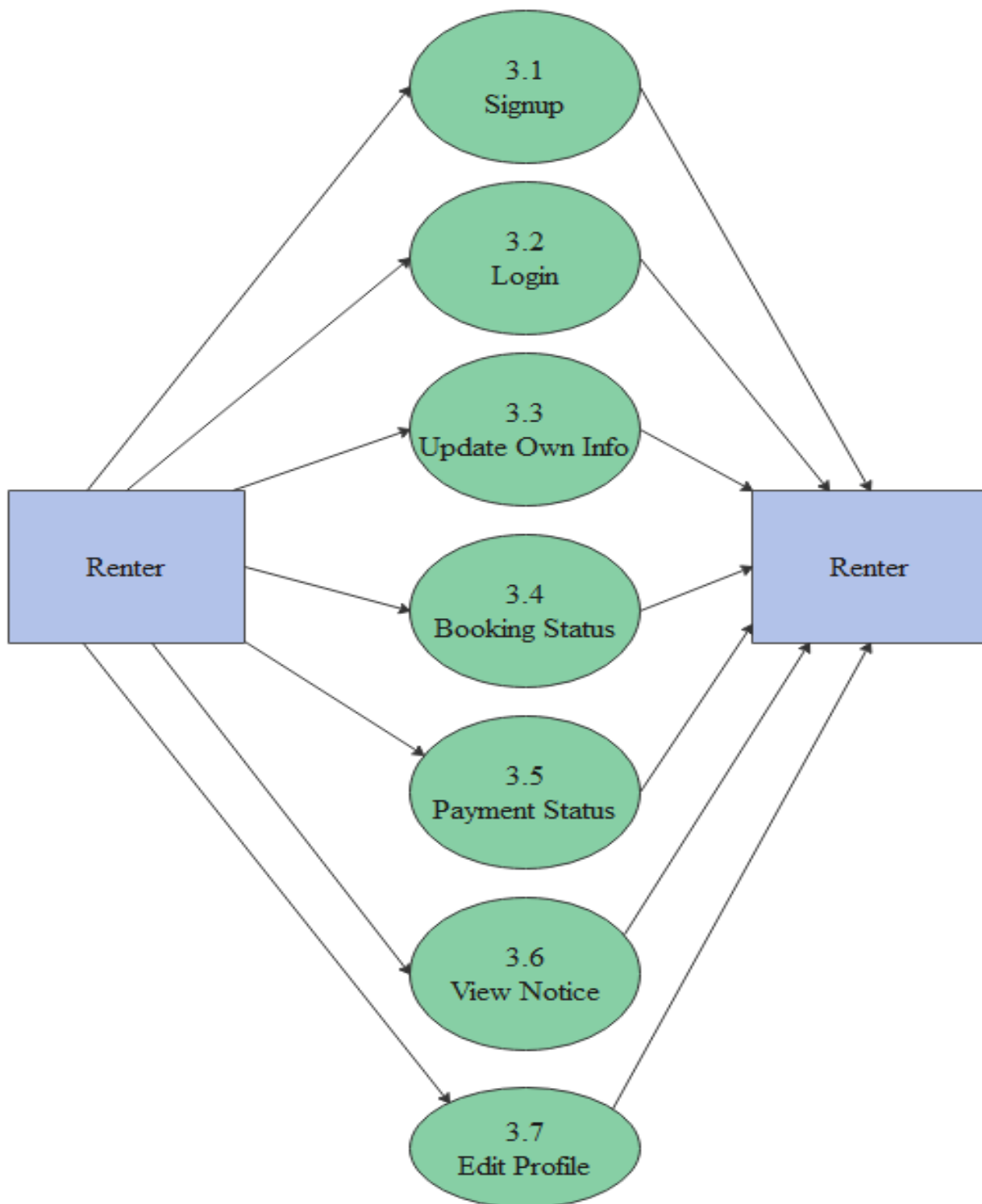


Figure 5.2.2.7: Level 1(Renter)

E. Level 1(Owner to Renter):

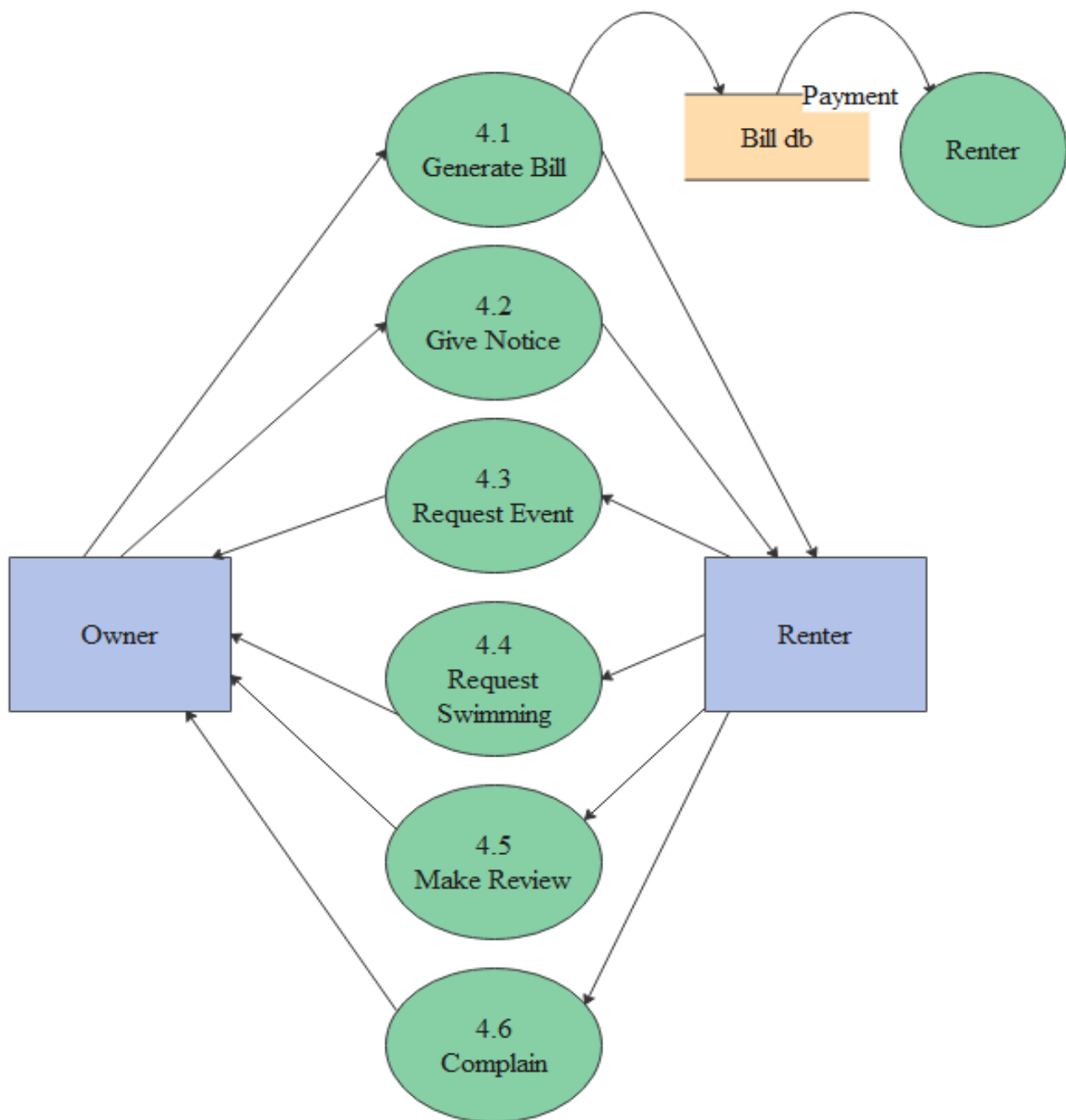


Figure 5.2.2.8: Level 1(Owner to Renter)

F. Level 2:

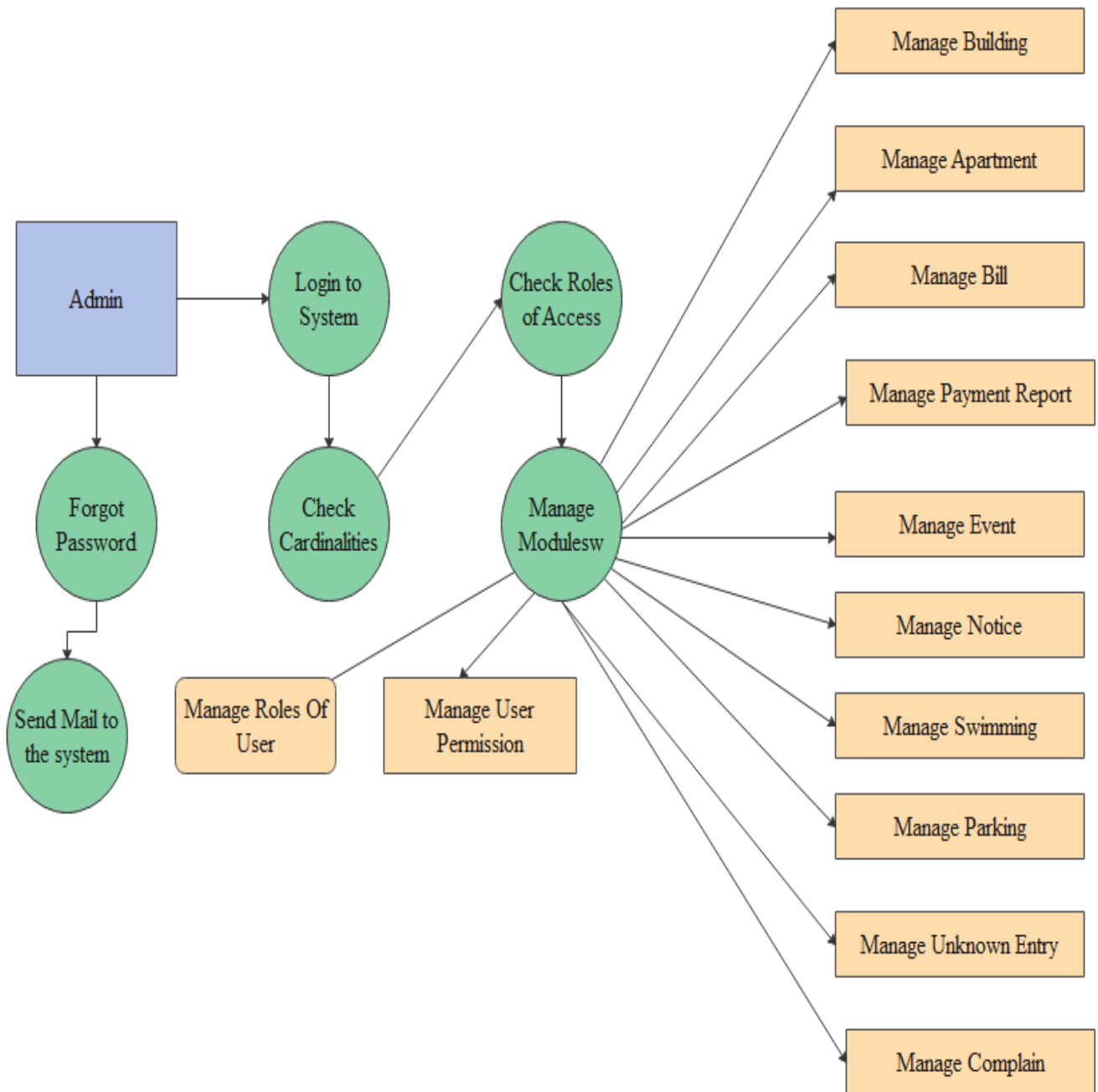


Figure 5.2.2.9: Level 2

5.3 Interface Design:

5.3.1 Home Page:

In our homepage a viewer can see the building by sliding, our services, latest news, latest apartment, testimonial, agents, and also our amazing properties.

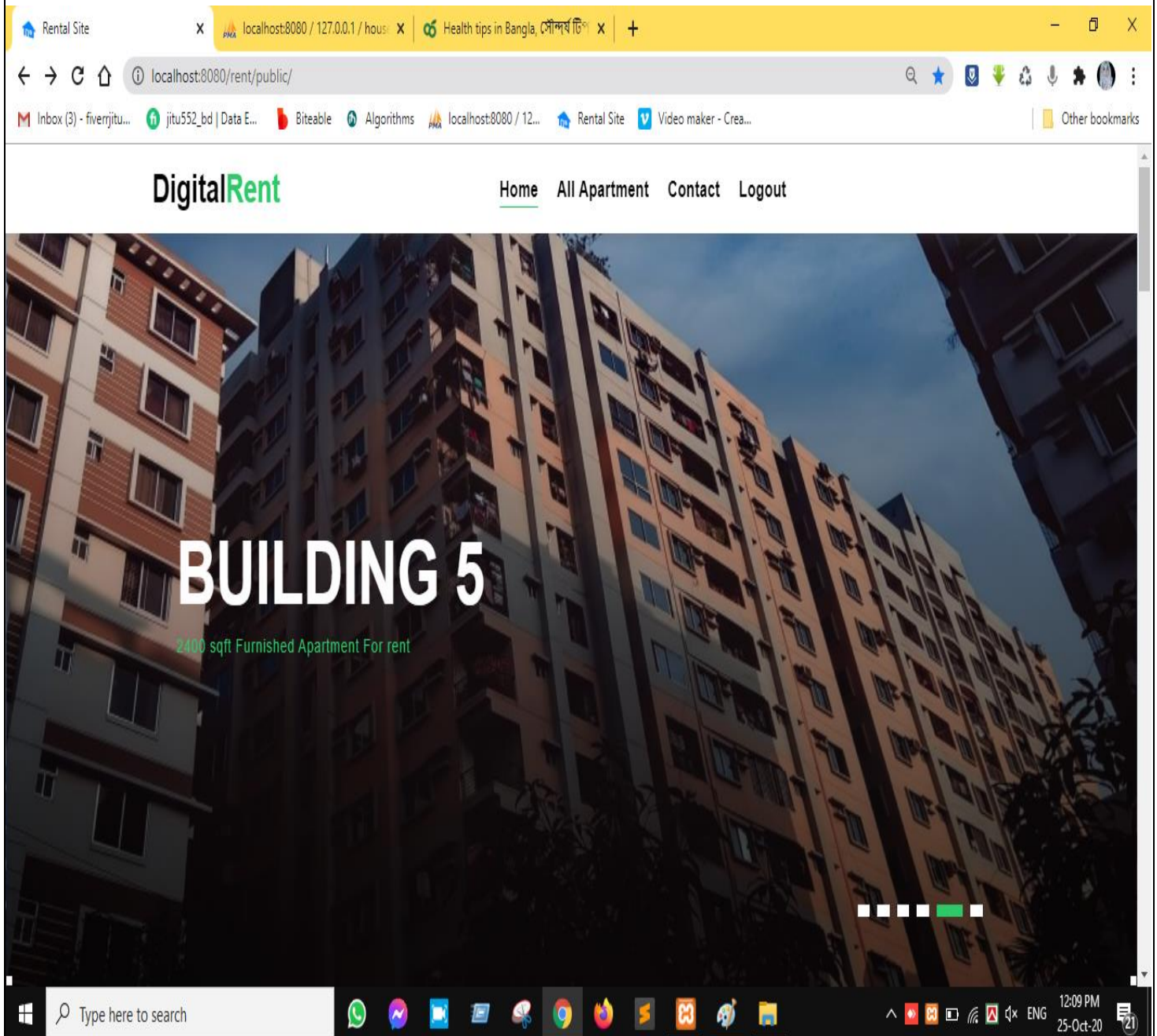


Figure 5.3.1.1: Building

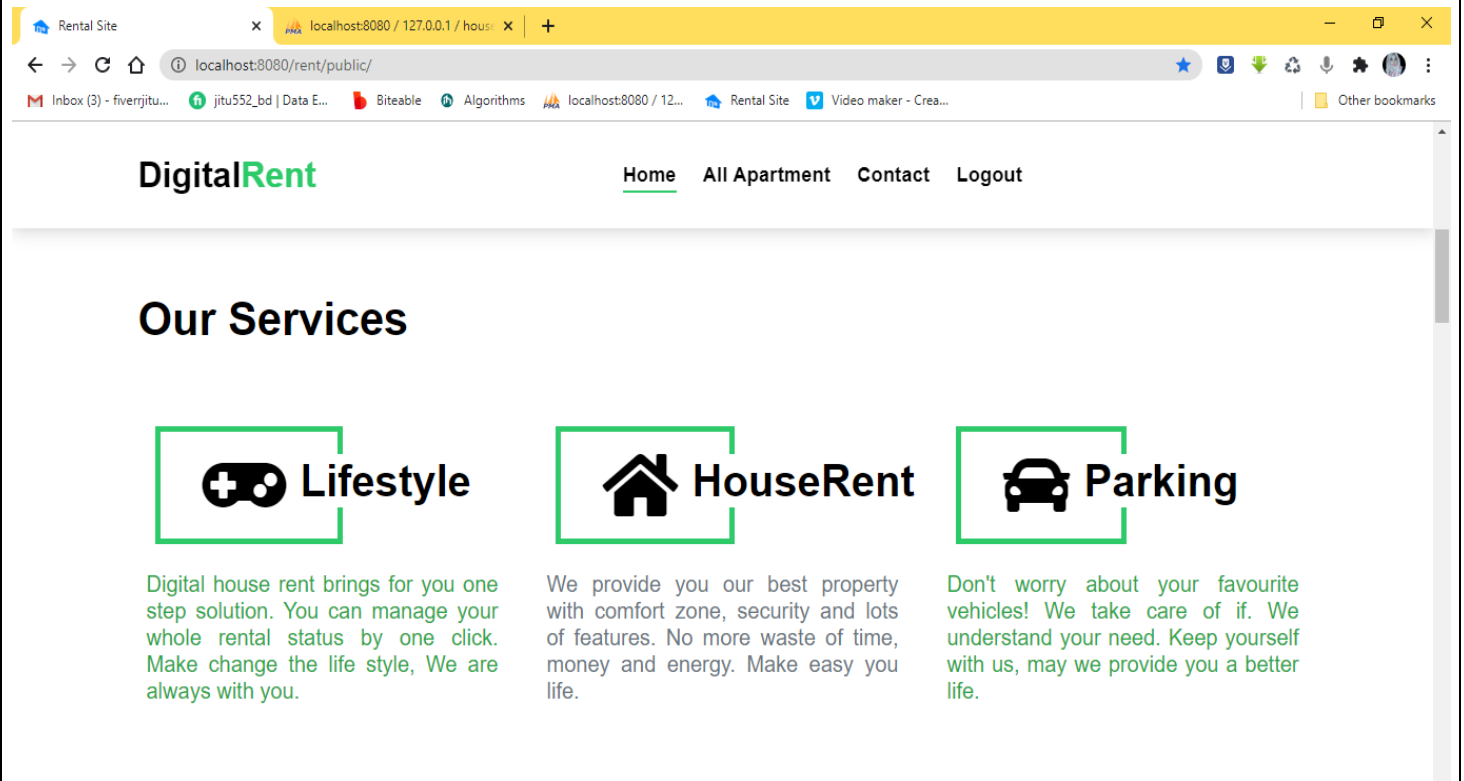


Figure 5.3.1.2: Our Services

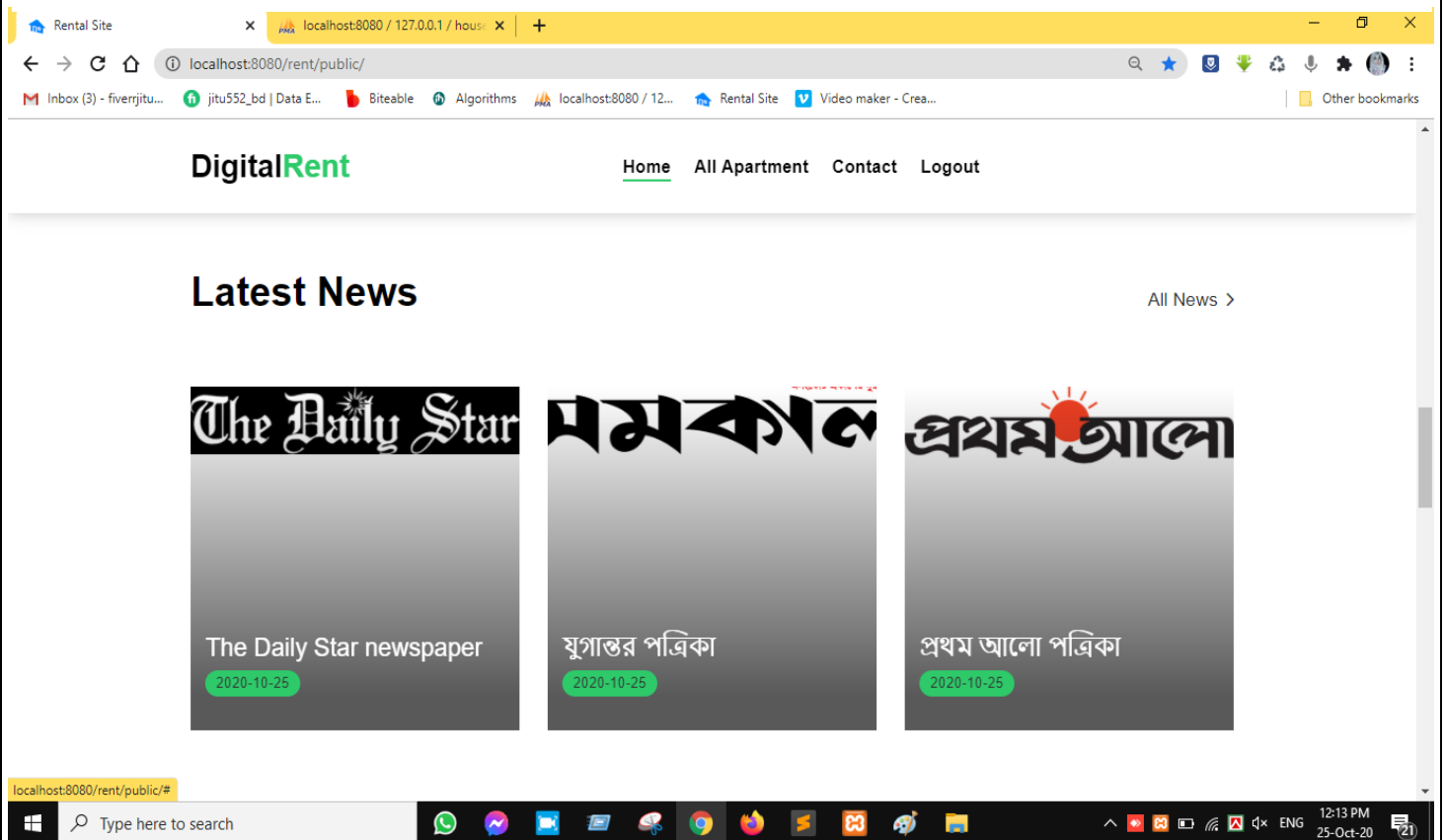


Figure 5.3.1.3: Latest News

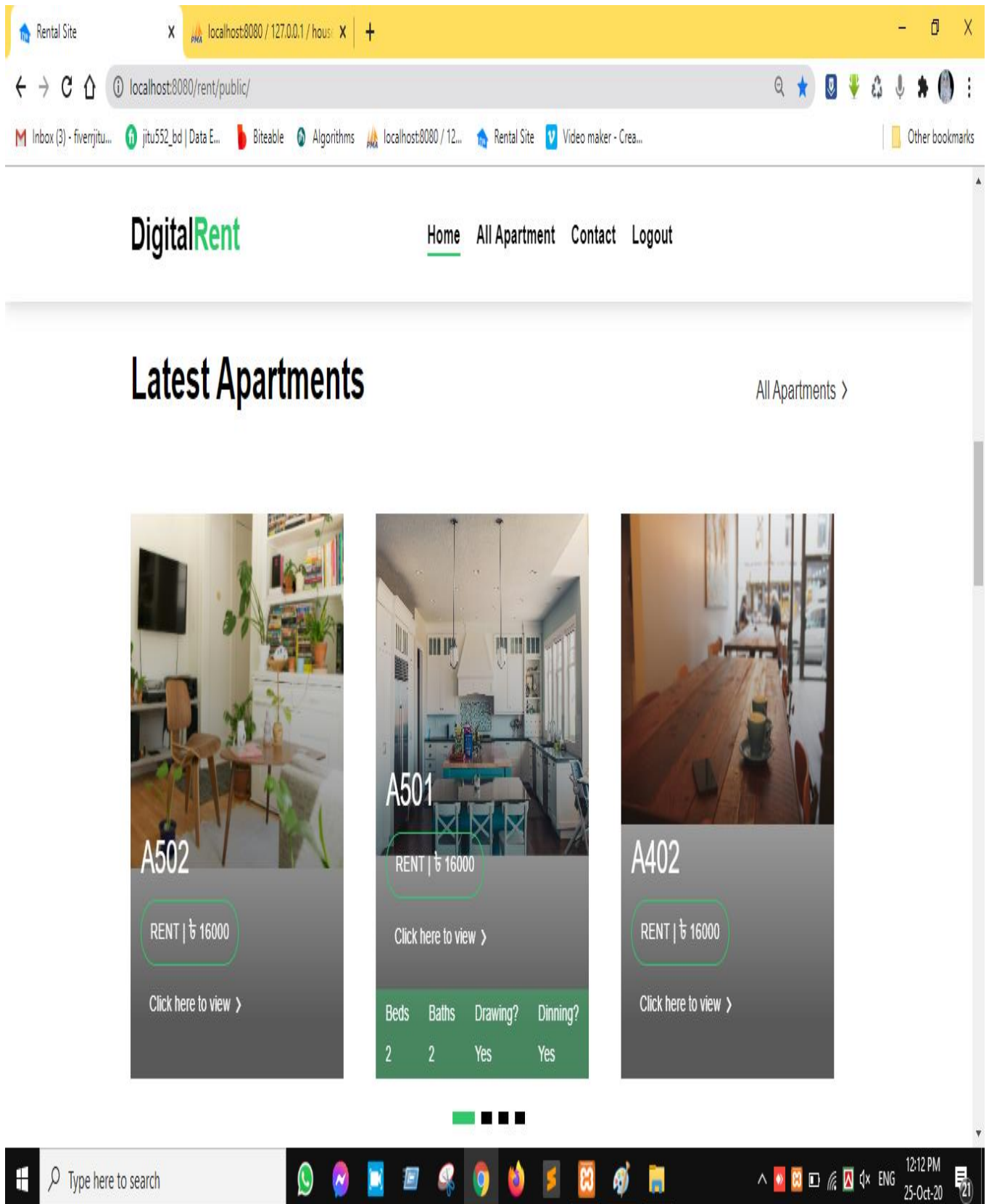


Figure 5.3.1.4: Latest Apartment

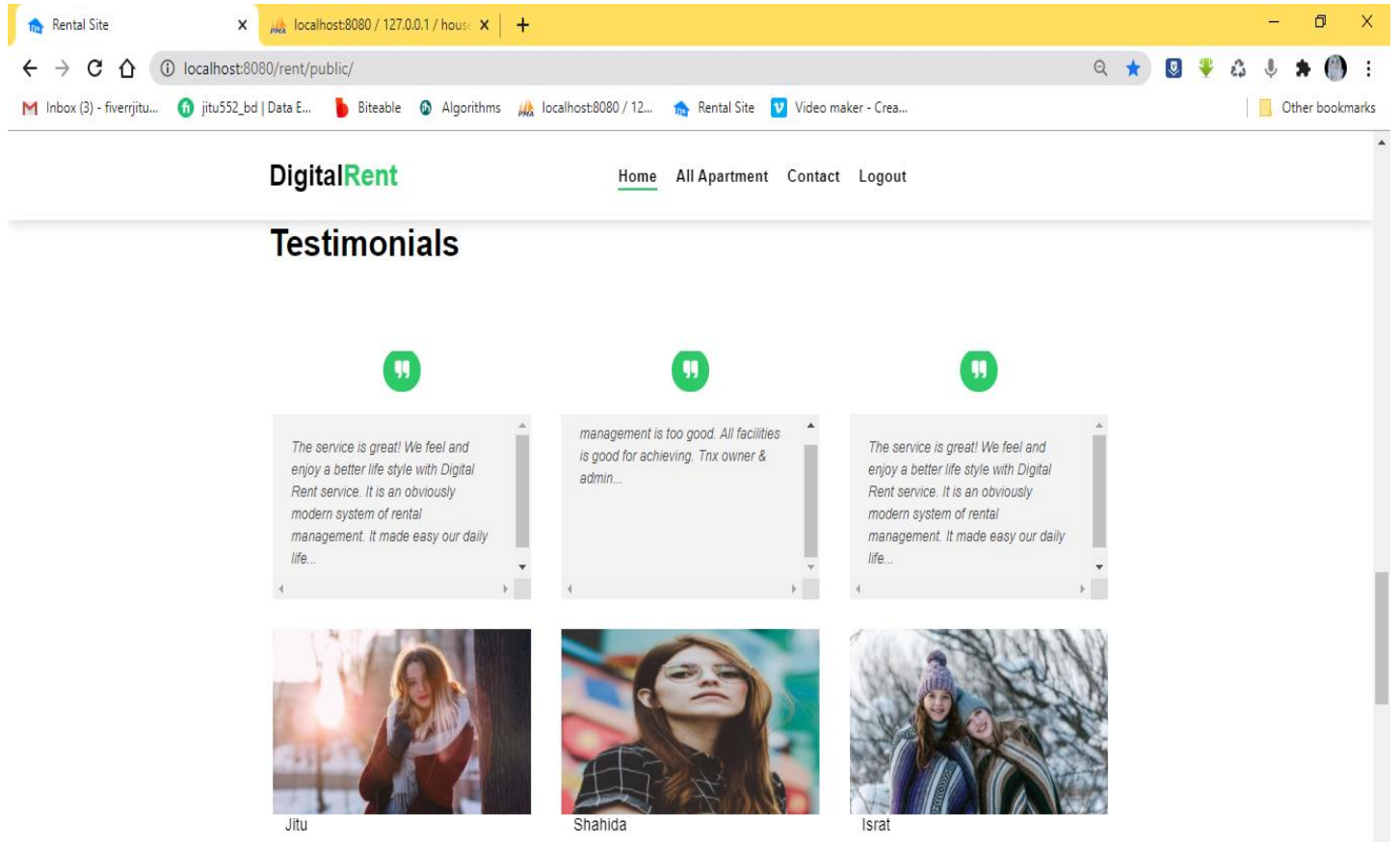


Figure 5.3.1.5 : Testimonials

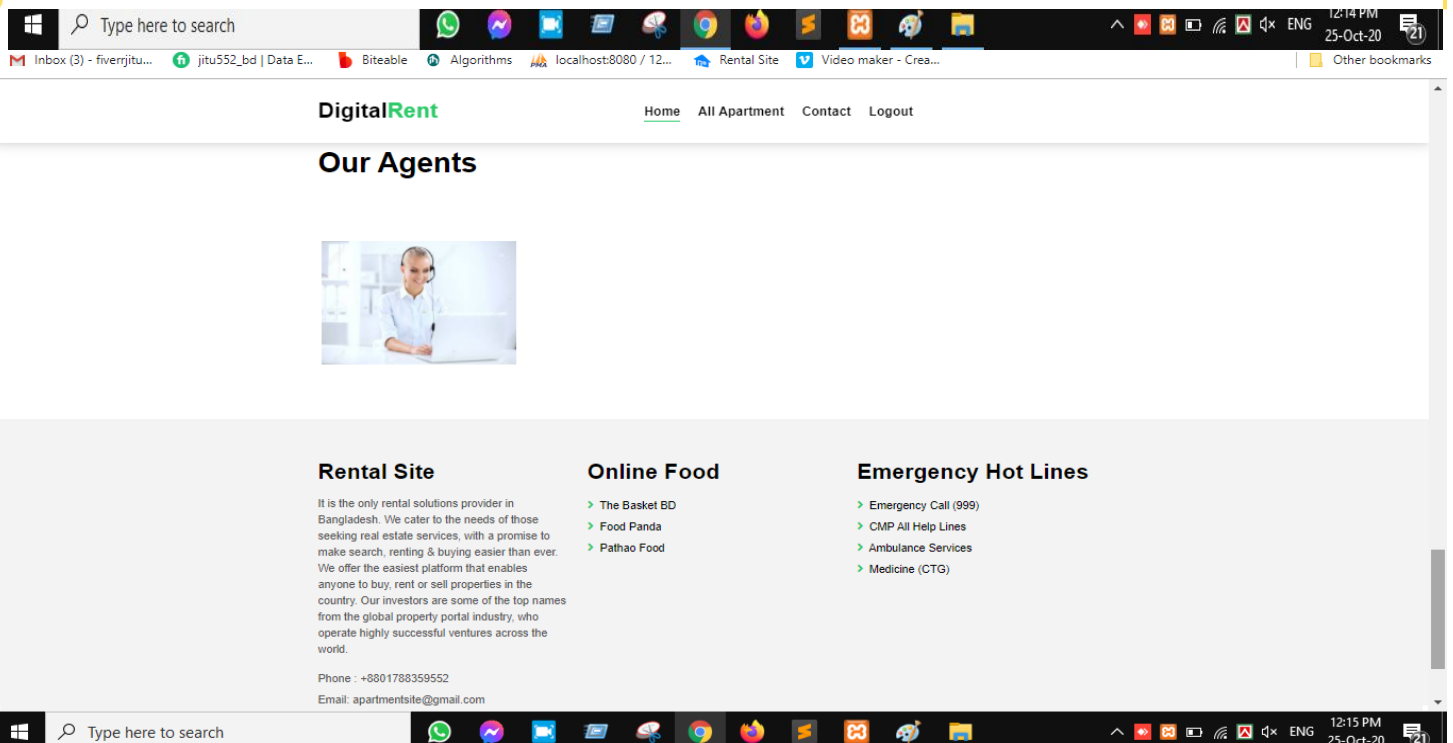


Figure 5.3.1.6: Our Agents

5.3.2 All Apartments:

In all apartment a renter or viewer can see the apartment no and the price of the apartment and by clicking here view viewer can see the apartment details and if he wants to book the apartment, he submits the booking request after that he fill-up the sign-up form. After filling up the sign-up form properly he must wait for the accepting his request.

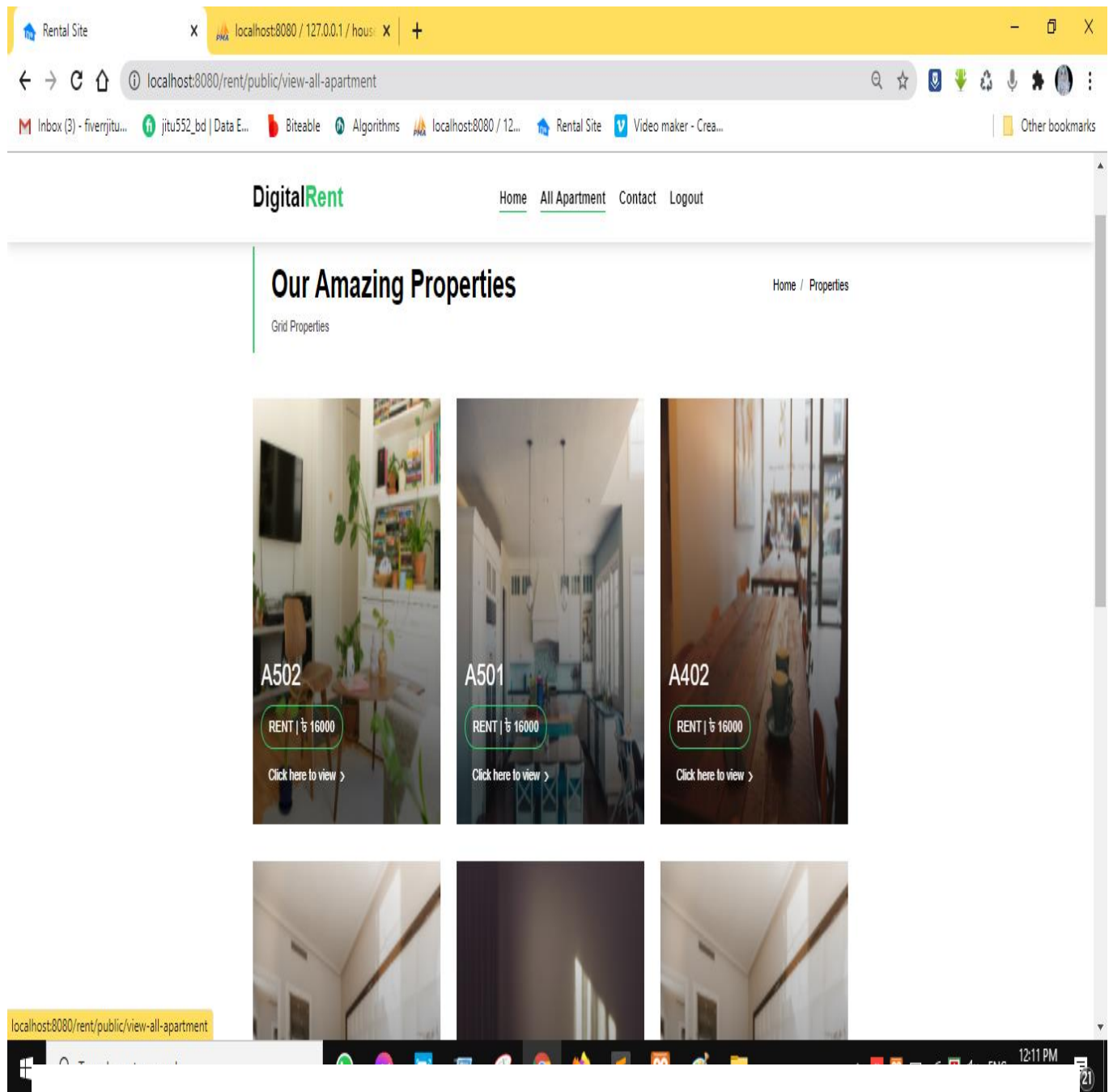


Figure 5.3.2: All Apartments

5.3.3 Signup Form:

After submitting the booking request a viewer must fill-up the sign-up form. After filling up the sign-up form properly he must wait for the accepting his request by owner.

Create User Account

Name ✓

Email ✓

Password ✓

NID/Birth Certificate ✓

Father's Name ✓

Mother's Name ✓

Mobile No ✓

Blood Group ✓

Image No file chosen ✓

Permanent Address ✓

[Create Account](#)

Figure 5.3.3: Signup Form

5.3.4 Login Page:

After accepting the request by owner renter can login by providing the valid password and email address.

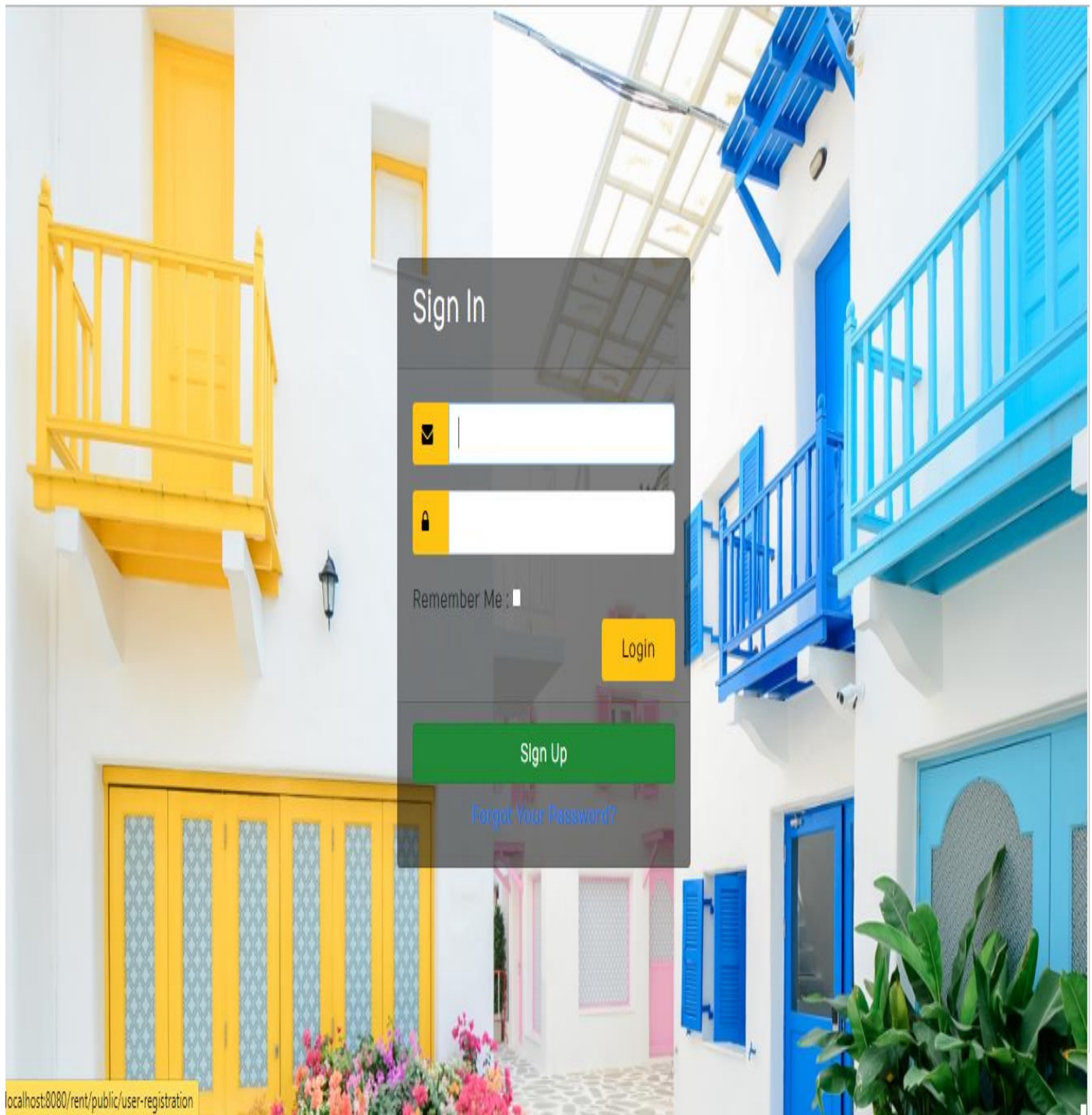


Figure 5.3.4: Login form

5.3.5 Owner Dashboard:

This is the owner dashboard where owner can manage all like Apartment allotment details, User list, Building, Billing list, Parking list, Event details etc. He can also add admin and watchman and pay their salary.

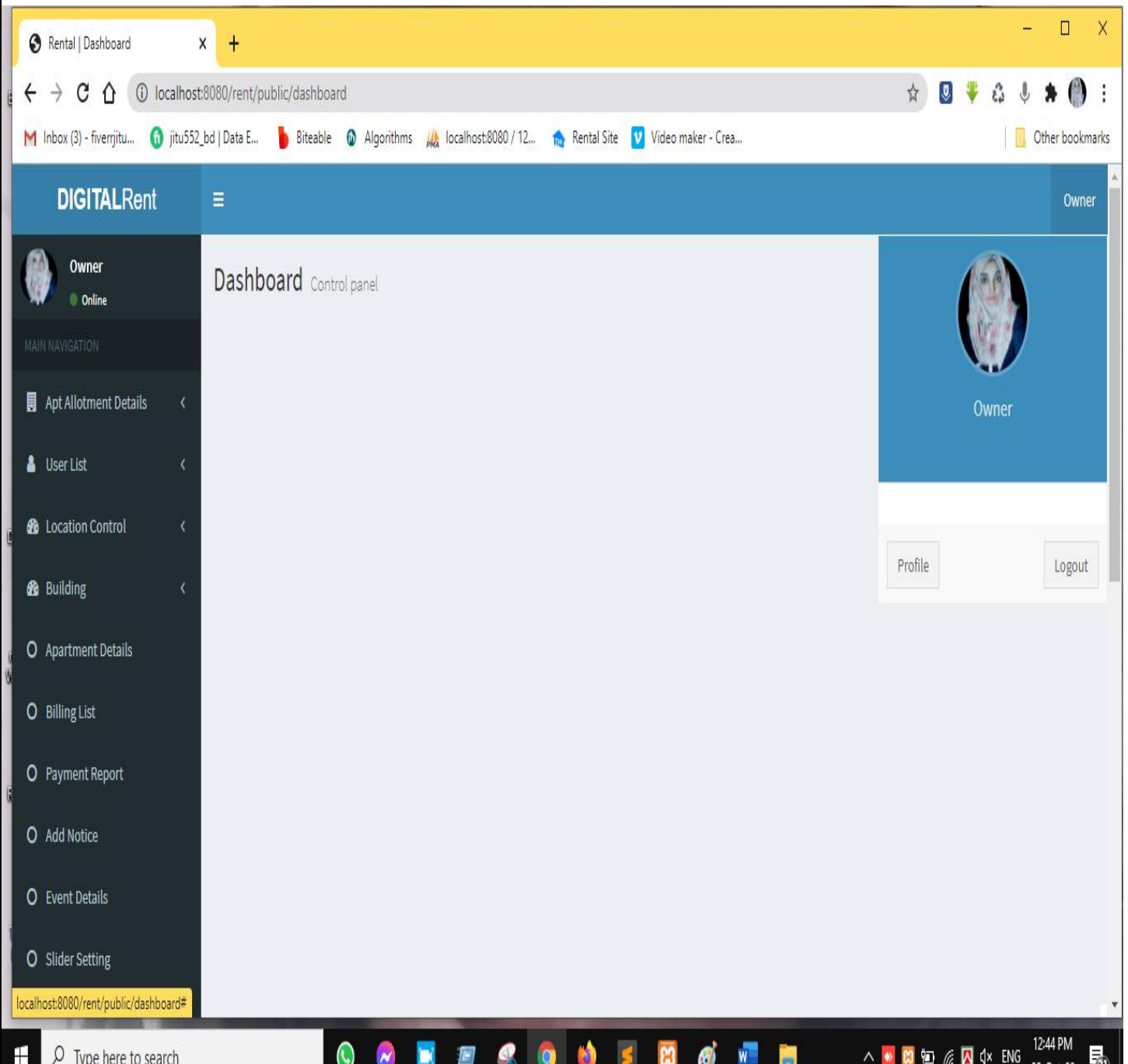


Figure 5.3.5: Owner Dashboard

5.3.6 Admin Dashboard:

This is the admin dashboard where owner can manage all like Apartment allotment details, User list, Building, Billing list, Parking list, Event details, swimming management etc. But he cannot add admin or watchman and cannot pay their salary.

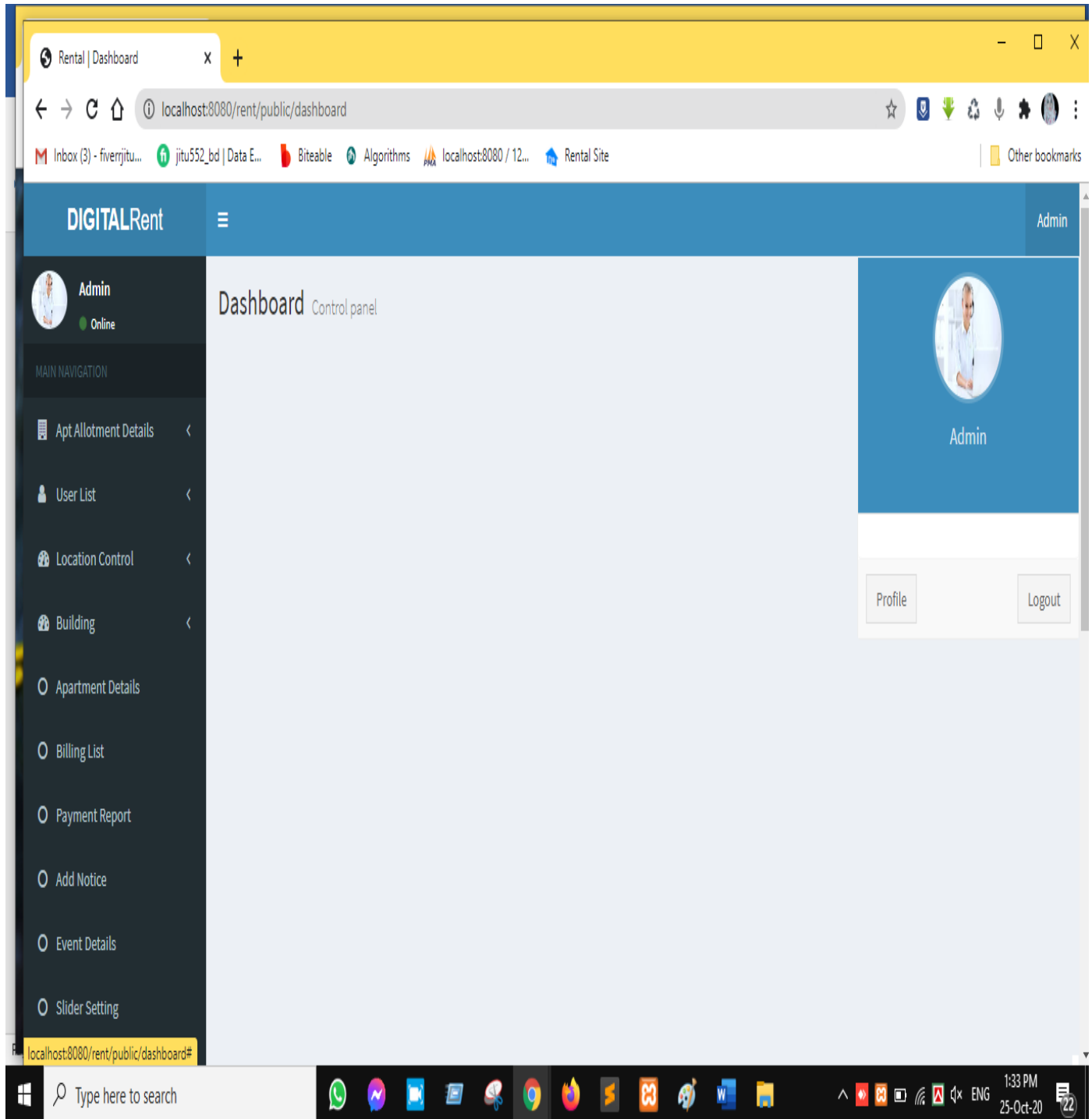


Figure 5.3.6: Admin Dashboard

5.3.6.1 Apartment Allotment Details:

In apartment allotment details there have two portions. Booking request and billing list. In booking request admin and Owner can see the request of renter, details of renter and if they want to accept they can accept it or deny it. After accepting the request both can see the renter's Name, Mobile, Advance, Booked form, dues, etc.

DIGITALRent Admin

Admin Online

MAIN NAVIGATION

- Apt Allotment Details
- Booking Request
- Booking List
- User List
- Location Control
- Building
- Apartment Details
- Billing List
- Payment Report
- Add Notice
- Event Details

Pending Requests

Requests

SL	Request By	Mobile	Father's Name	NID/Birth Certificate	Building	Apartment	Action
1	Jitu	01788359552	Motiur	7542369258931	Building 1	A501	<button>Accept</button> <button>Deny</button>

localhost:8080/rent/public/dashboard/request/booking-requests

Type here to search

9:48 AM 25-Oct-20

DIGITALRent

Admin

Admin

Online

MAIN NAVIGATION

Apt Allotment Details

Booking Request

Booking List

User List

Location Control

Building

Apartment Details

Billing List

Payment Report

Add Notice

Booking Status

Booking List

SL	Name	Mobile	Building	Apartment	Advance	Booked From	Dues	Action
1	Jitu	01788359552	Building 1	A102	20000/=	15-06-2020	-3600	Release
2	Shahida	01788953665	Building 1	A202	16000/=	15-02-2020	-16000	Release
3	Israt	01235896778	Building 1	A302	/=	15-03-2020	16400	Release

Figure 5.3.6.1: Apartment Allotment Details

5.3.6.2 User list:

In user list there is admin details, watchman details, Renter details. Both admin and renter can see the list. But only owner can add admin and watchman.

The screenshot displays the DIGITALRent application interface. The top header is blue with the 'DIGITALRent' logo on the left and a user profile 'Admin' on the right. The left sidebar is dark blue and contains a 'MAIN NAVIGATION' section with the following items: 'Apt Allotment Details', 'User List' (which is expanded), 'Admin Details', 'Watchman Details', 'Renter Details', 'Location Control', 'Building', 'Apartment Details', 'Billing List', 'Payment Report', and 'Add Notice'. The 'User List' item is expanded, showing a sub-menu with 'Admin Details', 'Watchman Details', and 'Renter Details'. The main content area is light blue and features an 'Add Admin' button at the top. Below the button is a table titled 'Admin List' with the following columns: SL, Name, Mobile, NID/Birth Certificate, Shift, Image, Salary, Father's Name, Mother's Name, and Address. The table contains one row of data for an admin user.

SL	Name	Mobile	NID/Birth Certificate	Shift	Image	Salary	Father's Name	Mother's Name	Address
1	Admin	018			Pro Pic	0	abc	xyz	

The address bar at the bottom of the browser window shows the URL: localhost:8080/rent/public/dashboard/userlist. The Windows taskbar at the very bottom shows the time as 9:52 AM on 25-Oct-20.

Figure 5.3.6.2: User List

5.3.6.3 Location Control:

Both admin and owner can access all cities, Thana and ward list. They can edit or delete any city, ward or Thana.

The screenshot shows the DIGITALRent web application interface. The top header is blue with the 'DIGITALRent' logo and a user profile 'Admin' with a status 'Online'. The left sidebar contains a 'MAIN NAVIGATION' menu with options: 'Apt Allotment Details', 'User List', 'Location Control' (expanded), 'All City' (selected), 'All Thana List', 'All Ward List', 'Building', 'Apartment Details', 'Billing List', 'Payment Report', and 'Add Notice'. The main content area is titled 'ALL city List' and features a green 'Add city' button. Below the title is a table labeled 'List of cities' with the following data:

SL	City Name	Latitude	Longitude	Action
1	Naokhali	22.824638	91.101733	Edit Delete
2	Mymensingh	24.743448	90.398384	Edit Delete
3	Sylhet	24.894802	91.869034	Edit Delete
4	Khulna	22.82	89.550003	Edit Delete
5	Dhaka	23.728783	90.393791	Edit Delete
6	Chittagong	22.3419	91.815536	Edit Delete

The bottom of the screenshot shows a Windows taskbar with various application icons and a system clock indicating 10:25 AM on 25-Oct-20.

Figure 5.3.6.3.1: All City List

MAIN NAVIGATION					
Apt Allotment Details					
User List					
Location Control					
All City					
All Thana List					
All Ward List					
Building					
Apartment Details					
Billing List					
Payment Report					
Add Notice					
localhost:8080/rent/public/thanas					
List of thanas					
SL	City	thana Name	Latitude	Longitude	Action
1	Naokhali	Begumganj	22.824638	91.101733	Edit Delete
2	Dhaka	Gulshan	23.728783	90.393791	Edit Delete
3	Dhaka	Mohammadpur	23.728783	90.393791	Edit Delete
4	Chittagong	Panclaiash	22.3419	91.815536	Edit Delete
5	Chittagong	Khulshi	22.3419	91.815536	Edit Delete
6	Chittagong	Halishahar	22.3419	91.815536	Edit Delete
7	Chittagong	Chandgaon	22.3419	91.815536	Edit Delete
8	Chittagong	Bayazid	22.3419	91.815536	Edit Delete

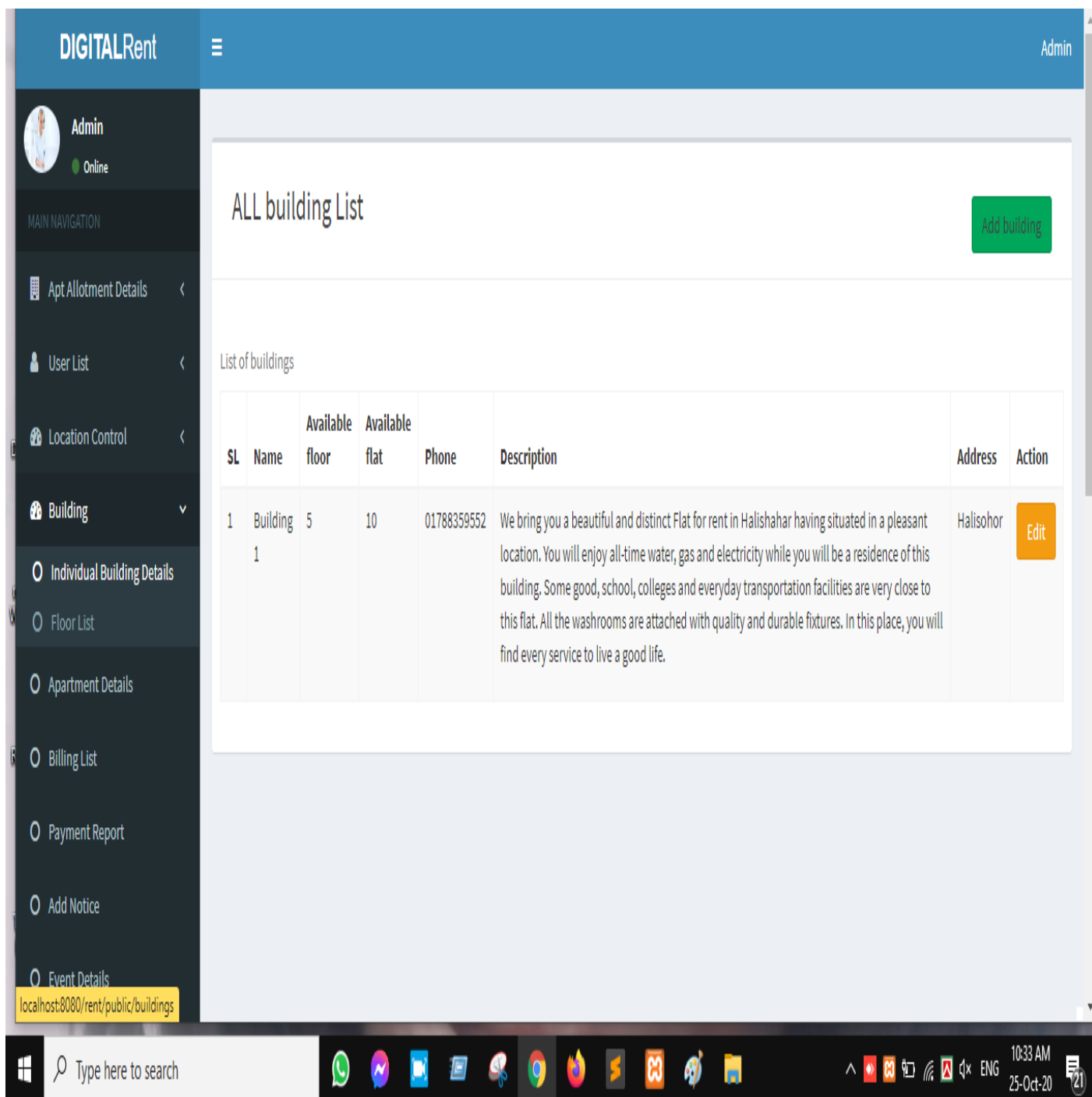
Figure 5.3.6.2: All Thana List

MAIN NAVIGATION					
Apt Allotment Details					
User List					
Location Control					
All City					
All Thana List					
All Ward List					
Building					
Apartment Details					
Billing List					
Payment Report					
Add Notice					
localhost:8080/rent/public/wards					
List of wards					
SL	City	ward Name	Latitude	Longitude	Action
1	Akbarshah	8	22.3419	91.815536	Edit Delete
2	Bayazid	2	22.3419	91.815536	Edit Delete
3	Chandgaon	9	22.3419	91.815536	Edit Delete
4	Halishahar	12	22.3419	91.815536	Edit Delete
5	Khulshi	5	22.3419	91.815536	Edit Delete
6	Panclaiash	5	22.3419	91.815536	Edit Delete
7	Mohammadpur	45	23.728783	90.393791	Edit Delete
8	Begumganj	5	22.824638	91.101733	Edit Delete

Figure 5.3.6.3: All Ward List

5.3.6.4 Building:

In the building list, there are two portions. Individual building details and floor list. The individual building details portion contains a name, available floor, available flat, phone, description, address. The floor list contains the building, floor name, available flat, etc.



The screenshot displays the DIGITALRent web application interface. The top header is blue with the 'DIGITALRent' logo and an 'Admin' user profile. The left sidebar contains a 'MAIN NAVIGATION' menu with options: 'Apt Allotment Details', 'User List', 'Location Control', 'Building' (selected), 'Individual Building Details', 'Floor List', 'Apartment Details', 'Billing List', 'Payment Report', 'Add Notice', and 'Event Details'. The main content area is titled 'ALL building List' and includes a green 'Add building' button. Below the title is a table labeled 'List of buildings'.

SL	Name	Available floor	Available flat	Phone	Description	Address	Action
1	Building 1	5	10	01788359552	We bring you a beautiful and distinct Flat for rent in Haliashahar having situated in a pleasant location. You will enjoy all-time water, gas and electricity while you will be a residence of this building. Some good, school, colleges and everyday transportation facilities are very close to this flat. All the washrooms are attached with quality and durable fixtures. In this place, you will find every service to live a good life.	Halisohor	Edit

The bottom of the image shows a Windows taskbar with various application icons and a system clock indicating 10:33 AM on 25-Oct-20.

Figure 5.3.6.4.1: Individual Building list

DIGITALRent

Admin

Admin

Online

MAIN NAVIGATION

Apt Allotment Details

User List

Location Control

Building

Individual Building Details

Floor List

Apartment Details

Billing List

Payment Report

Add Notice

Event Details

ALL floor List

Add floor

List of floors

SL	Building	floor Name	Available Flat	Action
1	Building 1	1st Floor	2	Edit Delete
2	Building 1	2nd Floor	2	Edit Delete
3	Building 1	3rd Floor	2	Edit Delete
4	Building 1	4th Floor	2	Edit Delete
5	Building 1	5th Floor	2	Edit Delete

Type here to search

WhatsApp

Telegram

Skype

Google

Firefox

Edge

File Explorer

10:34 AM

25-Oct-20

Figure 5.3.6.4.2 Floor list

5.3.6.5 Apartment Details:

In apartment details owner and admin access the list of building, floor, Apt no, room, bed, bath, dinning, drawing, price, measurement and if they want to edit anything, they can do that.

The screenshot displays the DIGITALRent Admin interface. The top header shows 'DIGITALRent' and 'Admin'. The sidebar on the left contains a user profile for 'Admin' (Online) and a 'MAIN NAVIGATION' menu with options: 'Apt Allotment Details', 'User List', 'Location Control', 'Building', 'Apartment Details' (selected), 'Billing List', 'Payment Report', 'Add Notice', 'Event Details', and 'Slider Setting'. The main content area is titled 'ALL apartment List' and features a green 'Add apartment' button. Below the title, it says 'List of apartments' and displays a table with the following data:

SL	Building	Floor	Apartment No	Room	Bed	Bath	Dinning	Drawing	Price	Cost	Measurement	Available	Description	Action
1	Building 1	1st Floor	A102	5	2	2	Yes	Yes	16000	400	2 Decimal	2020-01-01	We bring you a beautiful and distinct Flat for rent in Halishahar having situate.....	Edit
2	Building 1	3rd Floor	A302	5	2	2	Yes	Yes	16000	400	2 Decimal	2020-01-01	We bring you a beautiful and distinct Flat for rent in Halishahar having situate.....	Edit
3	Building 1	2nd Floor	A202	4	2	2	Yes	Yes	17000	400	2 Decimal	2020-01-01	We bring you a beautiful and distinct Flat for rent in Halishahar having situate.....	Edit
4	Building 1	1st Floor	A101	5	2	2	Yes	Yes	16000	400	2 Decimal	2020-01-01	We bring you a beautiful and distinct Flat for rent in Halishahar having situate.....	Edit

The bottom of the image shows a Windows taskbar with various application icons and a system clock indicating 10:35 AM on 25-Oct-20.

Figure 5.3.6.5: Apartment Details

5.3.6.6 Billing list:

Admin or owner can see the renter's name, mobile, building, apartment, total cost, previous dues, receive amount, current dues, advance etc.

DIGITALRent Admin

Billing List + Generate Bill

SL	Name	Mobile	Building	Apartment	Total Cost	Previous Dues	Receive Amount	Current Dues	Advance	Action
1	Israt	01235896778	Building 1	A302	16400 House Rent : 16000 WASA Bill : 400 Parking : 1000	0			+	Payment
2	Jitu	01788359552	Building 1	A102	16400 House Rent : 16000 WASA Bill : 400 Parking : 1000	0			+ 20000	Payment

Figure 5.3.6.6: Billing List

5.3.6.7 Payment report:

In payment report owner can see all payment of renter and report can be printed.

Apt.Allotment Details User List Location Control Building

Apartment Details

Billing List

Payment Report

Add Notice

Event Details

Slider Setting

Swimming Management

View Messages

View Complain

Update Company

News Links

Manage Linkup

Manage Health Tips

All Parking List

Unknown Entry

01-Jan-2020

to

12-Jan-2021

GO

Figure 5.3.6.7: Payment Report

5.3.6.8 Add Notice:

If owner want to give any notice for renter then he can give the notice-by-notice list. He can choose the building, floor, title and description of the renter.

The screenshot displays the DIGITALRent Admin interface. The top header is blue with the 'DIGITALRent' logo and a user profile 'Admin' with a green 'Online' status. The left sidebar contains a 'MAIN NAVIGATION' menu with options: 'Apt Allotment Details', 'User List', 'Location Control', 'Building', 'Apartment Details', 'Billing List', 'Payment Report', 'Add Notice', 'Event Details', and 'Slider Setting'. The main content area is titled 'ALL Notice List' and features a green 'Add Notice' button. Below the title is a table labeled 'List of Notices' with the following data:

SL	Building	Floor	Title	Description	Action
1	Building 1	1st Floor	Don't make too much sound..plz	For peace	Edit Delete
2	Building 1	2nd Floor	warning	Don't keep the garbage outside	Edit Delete

The bottom of the image shows a Windows taskbar with various application icons and a system tray indicating the time as 10:37 AM on 25-Oct-20.

Figure 5.3.6.8: Add Notice

5.3.6.9 Event details:

If any renter wants to arrange any event then he select event name, Date, Bill and admin accept or delete his request.

The screenshot shows the DIGITALRent Admin interface. The top header is blue with the DIGITALRent logo and a hamburger menu icon. The user profile 'Admin' is shown as online. The sidebar on the left contains a 'MAIN NAVIGATION' section with various options. The main content area is titled 'Event List' and contains a table with the following data:

SL	Event Name	Date	Food Menu	Bill	Status	Action
1	Birthday	24/10/20 -To- 24/10/20	Chicken Biryani, Chocolate cake, Soft drinks		Not Completed	Complete Delete

The browser address bar shows the URL: localhost:8080/rent/public/renter/event/view. The Windows taskbar at the bottom displays various application icons and the system clock showing 10:40 AM on 25-Oct-20.

Figure 5.3.6.9: Event List

5.3.6.10 Slider Setting:

Admin can add new slider, image and description.

Figure 5.3.6.10: Slider Setting

The screenshot displays the 'Slider Setting' page in a web application. The page is titled 'Add New Sliders' and includes a form with the following fields:

- Title:** A text input field.
- Description:** A text input field.
- Choose Slider Image:** A button labeled 'Choose File' with the text 'No file chosen' next to it.
- Save:** A blue button to submit the form.

Below the form is a table listing existing sliders:

SL	Header	Slider Image	Slider description	Action
1	Building 1		2400 sqft Furnished Apartment For rent	
2	building 2		2400 sqft Furnished Apartment For rent	
3	Building 3		2400 sqft Furnished Apartment For rent	
4	Building 4		2400 sqft Furnished Apartment For rent	
5	Building 5		2400 sqft Furnished Apartment For rent	

5.3.6.11 Swimming management:

In this portion Admin divide the day and time slot. He can also see the schedule list.

The screenshot shows the DIGITALRent Admin interface. The top navigation bar includes the logo and a user profile 'Admin' with an 'Online' status. The left sidebar lists various navigation options, with 'Swimming Management' expanded to show 'Add Schedule' and 'View Swimming List'. The main content area is titled 'View Schedule' and features a '+ Add Schedule' button. Below this is a 'Schedule List' table with the following data:

SL	Gender	Day	Time Range	Action
1	Female	Wednesday	8.00am to 10pm	Delete
2	Female	Wednesday	10:00 AM - 11:00AM	Delete

Figure 5.3.6.11: Add Schedule

Browser tabs: Rental | Dashboard, localhost:8080 / 127.0.0.1 / hous...

Address bar: localhost:8080/rent/public/swimming/swim-schedules#

Bookmarks: Inbox (3) - fiverrjitu..., jitu552_bd | Data E..., Biteable, Algorithms, localhost:8080 / 12..., Rental Site, Video maker - Crea..., Other bookmarks

DIGITALRent Admin

Admin Online

MAIN NAVIGATION

- Apt Allotment Details
- User List
- Location Control
- Building
- Apartment Details
- Billing List
- Payment Report
- Add Notice
- Event Details
- Slider Setting
- Swimming Management
 - Add Schedule
 - View Swimming List
 - View Messages

View Schedule

Schedule List

SL	Renter	Flat	Schedule For	Date	Day	Time Range	Action
1	Jitu	Building 1 (A102)	Female	01-01-2020	Wednesday	8.00am to 10pm	Delete

localhost:8080/rent/public/swimming/swim-schedules

Windows taskbar: 10:43 AM, 25-Oct-20

Figure 5.3.6.11.1: View Schedule

5.3.6.12 View message:

If renter sends any message then admin can see the message.

The screenshot shows a web application interface for an admin user. The top navigation bar includes the 'Rental | Dashboard' link and several active tabs: 'localhost:8080 / 127.0.0.1 / house...', 'Settings - Check passwords', and a plus sign for more tabs. The address bar shows the URL 'localhost:8080/rent/public/company/visitor-messages'. The sidebar on the left lists various navigation options, with 'View Messages' currently selected. The main content area is titled 'View Messages' and shows a breadcrumb trail '/ Admin / Messages'. Below this, a section titled 'Message' contains a table with the following data:

SL	Name	Phone	Email	Suggestion
1	Jitu	01788359552	jitu@gmail.com	There is a problem in room no 3 of 1No building

The Windows taskbar at the bottom shows the system time as 10:49 AM on 25-Oct-20.

Figure 5.3.6.12: View Message

5.3.6.13 Update company:

If admin want to update any company information then he can do it.

The screenshot displays the DIGITALRent Admin Dashboard. The top navigation bar is blue with the 'DIGITALRent' logo and an 'Admin' user profile. A sidebar on the left lists various management options. The main content area is titled 'Update Company' and contains a 'Profile Settings' section with several input fields for company information. Below this is a 'Social Setting' section for social media links. The browser's address bar shows the URL 'localhost:8080/rent/public/company/edit-company/1'.

Browser Tabs: Rental | Dashboard, localhost:8080 / 127.0.0.1 / hous..., Settings - Check passwords

Browser Address Bar: localhost:8080/rent/public/company/edit-company/1

Browser Bookmarks: Inbox (3) - fiverjitu..., jitu552_bd | Data E..., Biteable, Algorithms, localhost:8080 / 12..., Rental Site, Video maker - Crea..., Other bookmarks

Dashboard Header: DIGITALRent, Admin

Admin Profile: Admin, Online

MAIN NAVIGATION:

- Apt Allotment Details
- User List
- Location Control
- Building
- Apartment Details
- Billing List
- Payment Report
- Add Notice
- Event Details
- Slider Setting
- Swimming Management
- View Messages
- Update Company

Update Company Form:

Profile Settings

Company Name :	Phone 1 :	Phone 2 :	Mail Address 1 :
Modern Apartment Management	+8801788359552	+8801632350627	apartmentsite@gmail.com
Mail Address 2 :	About Company for Footer :		
rentalsite@gmail.com	It is the only rental solutions provider in Bangladesh. We cater to the needs of those seeking real estate services, with a promise to make search, renting & buying easier than ever. We offer the easiest platform that enables anyone to buy, rent or sell properties in the country. Our investors are some of the top names from the global property portal industry, who operate highly successful ventures across the world.		
Company Address :			
2No gate, CDA Avenue, East Nasirabad, Chattogram.			

Social Setting

Facebook:	Twitter:	whatsapp:	Instagram:
localhost:8080/rent/public/company/edit-company/1 om/rentalsite	https://twitter.com/rentalsite	01788359552	https://instagram.com/rentalsite

Figure 5.3.6.13: Update company

5.3.6.14 News Link:

Admin can update regular news from his dashboard by providing link.

The screenshot displays the DIGITALRent Admin Dashboard. The top header is blue with the 'DIGITALRent' logo on the left and 'Admin' on the right. A dark sidebar on the left contains a user profile for 'Admin' (Online) and a 'MAIN NAVIGATION' menu with items: 'Apt Allotment Details', 'User List', 'Location Control', 'Building', 'Apartment Details', 'Billing List', 'Payment Report', 'Add Notice', and 'Event Details'. The main content area is titled 'ALL NewsLink List' and includes a green 'Add NewsLink' button. Below this is a table titled 'List of NewsLinks' with columns: SL, Heading, Link, Date, Image, and Action. The table contains three entries, each with a 'Delete' button in the Action column. At the bottom right, there is a watermark for 'Activate Windows'.

SL	Heading	Link	Date	Image	Action
1	The Daily Star newspaper	https://www.thedailystar.net/	2020-10-25		Delete
2	যুগান্তর পত্রিকা	https://www.jugantor.com/	2020-10-25		Delete
3	প্রথম আলো পত্রিকা	https://www.prothomalo.com/	2020-10-25		Delete

Figure 5.3.6.14: News links

5.3.6.15 Manage Linkup:

In linkup list admin provide service provider's name , mobile ,address and web link.

The screenshot displays the DIGITALRent Admin dashboard. The top header is blue with the 'DIGITALRent' logo on the left and 'Admin' on the right. A dark sidebar on the left contains a user profile for 'Admin' (Online) and a 'MAIN NAVIGATION' menu with items: 'Apt.Allotment Details', 'User List', 'Location Control', 'Building', 'Apartment Details', 'Billing List', 'Payment Report', 'Add Notice', 'Event Details', and 'Slider Setting'. The main content area is titled 'Linkup List' and features a green 'Add Linkup' button. Below the title is a 'List of Link' table with the following data:

SL	Service Provider	Mobile	Address	Web Link	Action
1	Laundry Service	+880-31-611108 / +880-1787864188	321/A, Korbanigonj Chittagong, Bangladesh	http://thelaundryboybd.com/	Delete

An 'Activate Windows' watermark is visible in the bottom right corner of the main content area.

Figure 5.3.6.15: Manage Linkup

5.3.6.16 All Parking List:

In parking list admin can provide renter's name , building name , rent and vehicle no by which renter can select his parking slot.

DIGITALRent Admin

Admin
Online

MAIN NAVIGATION

- Apt Allotment Details
- User List
- Location Control
- Building
- Apartment Details
- Billing List
- Payment Report

ALL parking List

Add parking

List of parkings

SL	Name	Building Name	Rent	Vehicle No	Action
1	Jitu	Building 1	1000	1	<button>Edit</button> <button>Delete</button>
2	Shahida	Building 1	1000	1	<button>Edit</button> <button>Delete</button>
3	Israt	Building 1	1000	1	<button>Edit</button> <button>Delete</button>

Figure 5.3.6.16: All Parking

5.3.6.17 Unknown Entry:

In unknown entry list admin can indicate unknown person by selecting building, apartment, image, mobile, in time, description

DIGITALRent

Admin

Admin
Online

MAIN NAVIGATION

Apt Allotment Details

User List

Location Control

Building

Apartment Details

Billing List

Payment Report

Add Notice

ALL Entry List

Add Entry

List of apartments

SL	Building	Apartment	Image	Mobile	In time	Out time	Description	Action
1	Building1	A102		hj	01-01-1970 06:00 AM	01-01-1970 06:00 AM	ghjk	EditDelete
2	Building1	A302		01788359552	11-11-2020 11:50 PM	11-11-2020 11:50 PM	ghyg	EditDelete

Figure 5.3.6.17: Unknown Entry

5.3.6.18 Complain: If renter want to complain then he can complain to owner and owner see this

Owner
Online

MAIN NAVIGATION

- Apt Allotment Details
- User List
- Location Control
- Building
- Apartment Details
- Billing List
- Payment Report
- Add Notice

Complains

Complain List

SL	User Name	Building	Apartment	Complain	Date
1	Jitu	Building 1	A102	demo	11/11/20 01:15

Figure 5.3.6.18: Complain

5.3.7 Renter Dashboard: In renter dashboard renter can access all like booking status, payment bill, notice, event request, swimming schedule and he can also give review.

5.3.7.1 Renter Booking Status:

After accepting the booking request by owner renter can access the renter dashboard and see his name, mobile, building, apartment, advance, booked date.

The screenshot shows the DIGITALRent application interface. The top header is blue with the logo 'DIGITALRent' on the left and a user profile 'Jitu' on the right. The left sidebar is dark grey and contains a 'MAIN NAVIGATION' section with icons for 'Your Booking Status', 'Your Payment Status', 'Notice For You', 'Add Event Request', 'Add Swimming Schedule', and 'Make Review'. Below this is a 'LABELS' section with 'Important', 'Warning', and 'Information' indicators. The main content area is light blue and titled 'Booking Status'. It features a 'Billing List' table with the following data:

SL	Name	Mobile	Building	Apartment	Advance	Booked From
1	Jitu	01788359552	Building1	A102	20000/=	15-06-2020

At the bottom of the screen, a Windows taskbar is visible with the search bar, taskbar icons, and system tray showing the date '25-Oct-20' and time '12:11'.

Figure 5.3.7.1: Renter Booking status

5.3.7.2 Renter Payment Status:

In this section renter can see his own payment details like house rent, wasa cost and parking.

DIGITALRent

Jitu

Jitu
Online

MAIN NAVIGATION

Your Booking Status

Your Payment Status

Notice For You

Add Event Request

Add Swimming Schedule

Make Review

Complain

LABELS

Important

Warning

Information

Payment Details

Payment Transaction

Payments

SL	House Rent + WASA Bill + Parking	Billing Month/Date	Paid Amount	Dues	Payment Date
1	16400	June (29/06/20)	16400	Paid	08 July, 20
2	16400	July (30/07/20)	16400	Paid	18 August, 20
3	16400	August (27/08/20)	16400	Paid	04 January, 21
4	16400	September (24/09/20)	Not Paid	16400 Make Payment	-
5	16400	October (24/10/20)	Not Paid	16400 Make Payment	-
6	16400	November (19/11/20)	16400	Paid	05 December, 20
7	16400	December (19/12/20)	Not Paid	16400 Make Payment	-
Total Amount : 114800/=		Total Paid : 65600/=		Total Dues(Without Adding Advance) : 49200/=	

localhost:8080/rent/public/renter/renter-payment-status

Transact Payment - SSLCommerz

Your Info

Full name

Jitu

Mobile

01788359552

Email

shahidajitu552@gmail.com

Your Payment Info

24 September, 2020

House Rent

16000

Monthly House Rent

WASA Cost

400

Monthly WASA Cost

Parking

5000

Monthly Parking Cost

Total (BDT)

16400

Pay Now

The screenshot displays a payment gateway interface with a 'DEMO' watermark. At the top, there are navigation links: Support, FAQ, Offers (with a notification badge), and Login. Below these are three main sections: CARDS, MOBILE BANKING, and NET BANKING. The 'CARDS' section is active, showing logos for VISA, Mastercard, and American Express, followed by 'Other Cards'. The form fields include: 'Enter Card Number', 'MM/YY', 'CVC/CVV', and 'Card Holder Name'. There is a checkbox for 'Save card & remember me' and a link to 'Terms of Service'. At the bottom, a large button indicates the payment amount: 'PAY 16,400 BDT'.

Figure 5.3.7.2: Renter Payment Status

5.3.7.3 Notice for renter:

In this section renter can see the notice which is given by owner.

The screenshot displays the DIGITALRent web application interface. The top header is blue with the 'DIGITALRent' logo on the left and the user name 'Jitu' on the right. A sidebar on the left contains a user profile for 'Jitu' (Online) and a 'MAIN NAVIGATION' menu with options: 'Your Booking Status', 'Your Payment Status', 'Notice For You' (selected), 'Add Event Request', 'Add Swimming Schedule', and 'Make Review'. Below the navigation menu are 'LABELS' for 'Important' (red), 'Warning' (orange), and 'Information' (blue). The main content area is titled 'ALL Notice List' and shows a 'List of Notices' table. The table has columns for SL, Building, Floor, Title, Description, and Action. One notice is listed with SL 1, Building 1, 1st Floor, Title 'Don't make too much sound..plz', Description 'For peace', and a 'View' button in the Action column. The browser's address bar shows 'localhost:8080/rent/public/notice/renter'. The Windows taskbar at the bottom includes a search bar, various application icons, and system tray information showing the time as 12:05 PM on 25-Oct-20.

SL	Building	Floor	Title	Description	Action
1	Building 1	1st Floor	Don't make too much sound..plz	For peace	View

Figure 5.3.7.3: Notice for renter

5.3.7.4 Add Event Request:

If renter wants to arrange any events then he can add event name, date, food menu, and bill. If admin accept his booking then he can see the status which is completed or not completed.

DIGITALRent Jitu

Add Event + Add Event

Event List

SL	Event Name	Date	Food Menu	Bill	Status	Action
1	Birthday	24/10/20 - To - 24/10/20	Chicken Biryani, Chocolate cake, Soft drinks		Not Completed	Edit Delete

Labels: Important, Warning, Information

Windows Taskbar: Type here to search, 12:05 PM, 25-Oct-20

Figure 5.3.7.4: Add Event Request

5.3.7.5 Add Swimming schedule:

If renter want to swim then he can add swim schedule. If swimming schedule is available then it is accepted by admin.

DIGITALRent Jitu

View Schedule + Add Schedule For Swimming

Schedule List

SL	Schedule For	Date	Day	Time Range	Action
1	Female	01-01-2020	Wednesday	8.00am to 10pm	Delete

localhost8080/rent/public/renter/swimming

Type here to search

12:05 PM 25-Oct-20

Figure 5.3.7.5: Add Swimming Schedule

5.3.7.6 Make Reviews:

If renter want to review the apartment, he can give review of the apartment.

DIGITALRent Jitu

Review + Add Review

Review List

SL	Apartment	Display Name	Star	Review/Testimonial	
1	A102	Jitu	5	The service is great! We feel and enjoy a better life style with Digital Rent service. It is an obviously modern system of rental management. It made easy our daily life...	Delete

MAIN NAVIGATION

- Your Booking Status
- Your Payment Status
- Notice For You
- Add Event Request
- Add Swimming Schedule
- Make Review

LABELS

- Important
- Warning
- Information

Waiting for mc.yandex.ru...

Type here to search

12:06 PM 25-Oct-20

Figure 5.3.7.6: Make Review

5.3.7.7 Make Complain: If renter face any problem then he can give the complain

The screenshot displays a web application interface for a user named 'Israt' who is 'Online'. The sidebar on the left contains a 'MAIN NAVIGATION' menu with the following options: 'Your Booking Status', 'Your Payment Status', 'Notice For You', 'Add Event Request', 'Add Swimming Schedule', 'Make Review', and 'Complain'. The main content area is titled 'Review' and features a '+ Add Review' button in the top right corner. Below the title, there is a 'Review List' section containing a table with the following data:

SL	Complain	Date	Delete
1	I face some problem of washroom	13-11-2020 09:20	Delete

Figure 5.3.7.7: Make Complain

5.4 Work Plan and Timeline:

Weeks	Work Plan	
1	Pre- Project	
2	Feasibility Study	
2	Business study	
5	Functional model	Agree plan
		Identify functional prototype
		Review functional prototype
		Create functional prototype
		testing
5	Design And build	Identify design prototype
		Review design prototype
		Create design prototype
5	Implementation	Implement
		Train users
		User approval
		Review business aspect

Table 5.4 Workplan And Timeline Table

5.5 Implementation:

Converting system specification into and executable system to developing software. this is the process that occur in implementation stage. implementation of the software that can read the software design and coding. many software about apartment management system to gather many information for implementation. many software is created to maintain the meant but we have tried to add being of the user relational database to store and retrieve data for user which makes searching more reliable and easier.

5.3 Summary:

We have described the data flow diagram, activity diagram and showed software interface design at the end of the chapter anyone can get the idea about what is the relation between the system design and requirement engineering and how it affects design techniques.

CHAPTER-6:

6. TESTING & SECURITY

6.1 Introduction:

The process of test management in traditional development methods has a life cycle that is well-defined and stabilized by literature and adopted by any software company. However, when it comes to a process for the **Agile Testing** scenario, it is not the same. Some testing approaches in agile methods are found in literature to include them in agile development teams; however, these approaches are isolated actions of researchers and not a formal process of how to use the tests in agile methods, as occurs in traditional tests. In this article, we present an **Extreme Level Programming (XP)** research on deployment approaches of agile testing, successfully used in several contexts in agile methods, in order to identify the characteristics that have been used in this implementation process. The experiment aimed to analyze the application of the characteristics common to the test deployment approaches so that in future work it will be possible to define a formal process for deployment testing in agile methods.

6.2 Research Objective & Development Approach of Agile Methods:

The exploratory research of this study aimed to obtain information which help answer the following questions:

- 1) What agile testing deployment approaches are being used by researchers and software company?
- 2) What is the formal process used to insert testing in agile development teams?

Based on the surveyed studies it is possible to observe some characteristics of the testing deployment approaches used in a development based on agile methods types of extreme programming. The insertion of regression tests is a useful way to ensure that, as new features are added to software, the entire system is tested, to verify if a new feature does not affect the behavior of another.

Characteristics of Testing:

- A. Automation of Testing
- B. Integration Testing
- C. Continuous Integration Testing
- D. Tester as A Team Member

A. Test Automation:

For the automation of tests, the experiment was divided into three phases.

First Phase: Tool selection for test automation. The tool used was Visual Studio Test Professional because it is a tool that was already available in the company where the experiment was done.

Among the features of the tool are: i) the management of testing cases; ii) manual test run; and iii) recant-play tests, which consist of using graphic user interface tools to write and run testing cases, simulating then a user interacting with the application.

Second Phase: Monitoring of automated testing. The follow-up was performed by the tester and the emphasis in this phase was on the realization of automated testing using the recant-play feature, called Coded UI Test.

Third Phase: Results Analysis. With the use of the tool adopted it was observed that the construction of a Coded UI Test took an average of 45 minutes. The average number of defects found during the construction of Coded UI Test was 1.4 defects by functionality. In more complex features, i.e. which have a larger set of business rules and validations to be made, it was possible to identify an average of 3 different defects, as in less complex features, e.g. simple signup interfaces, no defect was found. Eight features were tested.

B. Regression Tests:

The team tester was responsible for running regression tests for each new feature. With the application of regression tests, it was possible to identify the incidence of defects in functionalities already tested and which have been affected by the new functionality. The tester reported the problem generated by the new functionality to the development team and the defect was registered.

C. Continuous Integration:

To enable continuous integration was also a role of the team tester. With each new version released, the tester updated the production environment, i.e. an environment different from that of development and which contained the latest version delivered to the customer.

D. Tester as A Team Member:

It is important to remark that the tester was part of the team and was responsible for monitoring the testing activities within the Sprint.

6.3 XP and Web Development:

XP's focus on small teams lets it replace paper-based documentation with face-to-face communication. Hence, it's a good fit for many Web based software projects, which often postpone documentation efforts because of time-to-market constraints. In XP, all developers work closely together so they can communicate informally rather than spending time documenting designs and decisions.

6.4 XP Practices:

XP proposes a set of software development practices to increase productivity while maintaining quality. We group XP's 12 practices according to three key areas: customer satisfaction, software quality, and development process organization.

A. Customer Satisfaction:

Building a high-quality software system is irrelevant if it does not solve the customer's problem. To increase customer satisfaction, XP uses one practice: small releases.

B. Small Releases:

Given that requirements change often, XP keeps release cycles short and ensures that each release produces a useful software system that generates business value for the customer. Short cycles reduce customer risk, letting the customer quickly terminate projects that fail to deliver business value. A short release cycle also helps developers deal with changing requirements and reduces the impact of planning errors.

C. Software Quality:

XP employs various practices to keep software quality high. Although some might appear unusual, their combined effects ensure that the team maintains high quality without slowing down the development process.

D. Metaphor:

A metaphor represents a coherent view of the system that makes sense to both the business and technical sides and represents "what we are trying to do." The metaphor is sometimes embodied in a single user story that portrays this idea and gives everyone the system basics. In a sense, the metaphor serves as the high-level software architecture.

6.5 XP Testing:

Software testing — specifically, automated regression testing—is a key part of XP. The customer defines functional (acceptance) tests, which the development team implements. From a business perspective, these tests verify that the program does what it is supposed to do. According to the XP philosophy, a feature lacking automated tests does not exist. In XP, programmers write unit tests before they write actual code. Writing test drivers before writing the code forces software developers to think about the problem before programming. The test drivers thus serve as a detailed

specification of the method's functionality. This is also one of the intentions of up-front design. The XP team can also use test drivers later to see if the system exhibits the proper functionality after the code is written. They can't do this with paper-based documentation.

Types of Testing:

- ☐ White box testing.
- ☐ Black box testing.

6.5.1 White Box Testing:

White box testing is based on internal structure or workings of a programmed as post to the functionality exposed to end user internal software and cold working should be known for this testing.

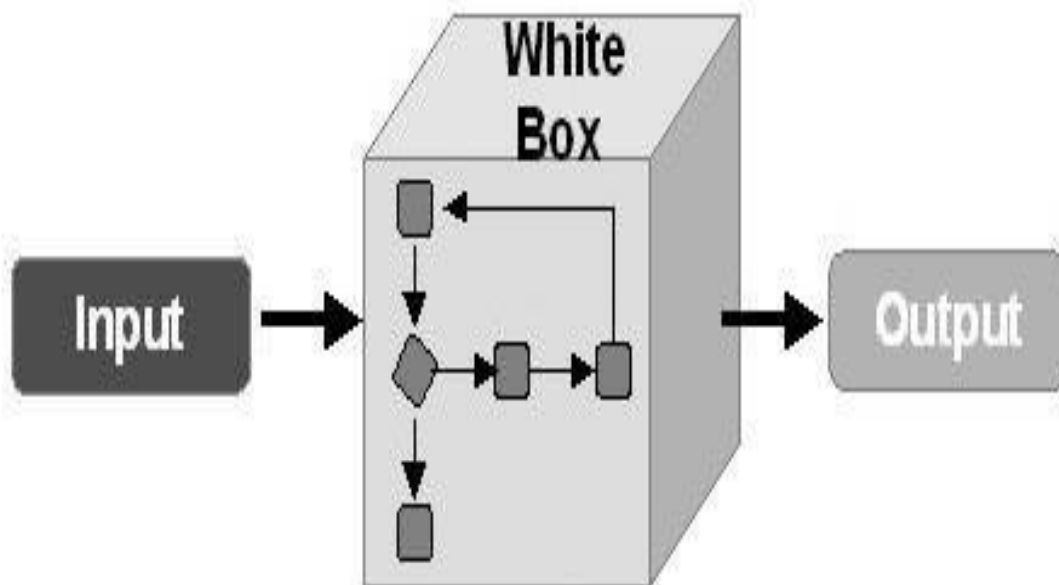


Figure 6.5.1: White Box Testing

A. Unit Tests:

Tests written from the programmer's perspective to identify possible system malfunctions. The team keeps all unit tests running all the time: Nothing is integrated into the baseline if a unit test fails. If a bug is encountered, the team first writes a unit test that shows the problem, and then fixes it.

6.5.2 Black Box Testing:

Black box testing out is a software testing approach in which functionality of the software below take a look at (SUT) is tested without looking at the inner code structure, implementation details, and understanding of inner paths of the software program. This kind of trying out is based absolutely on the software necessities and specs. It is completed by way of the subsequent tests:

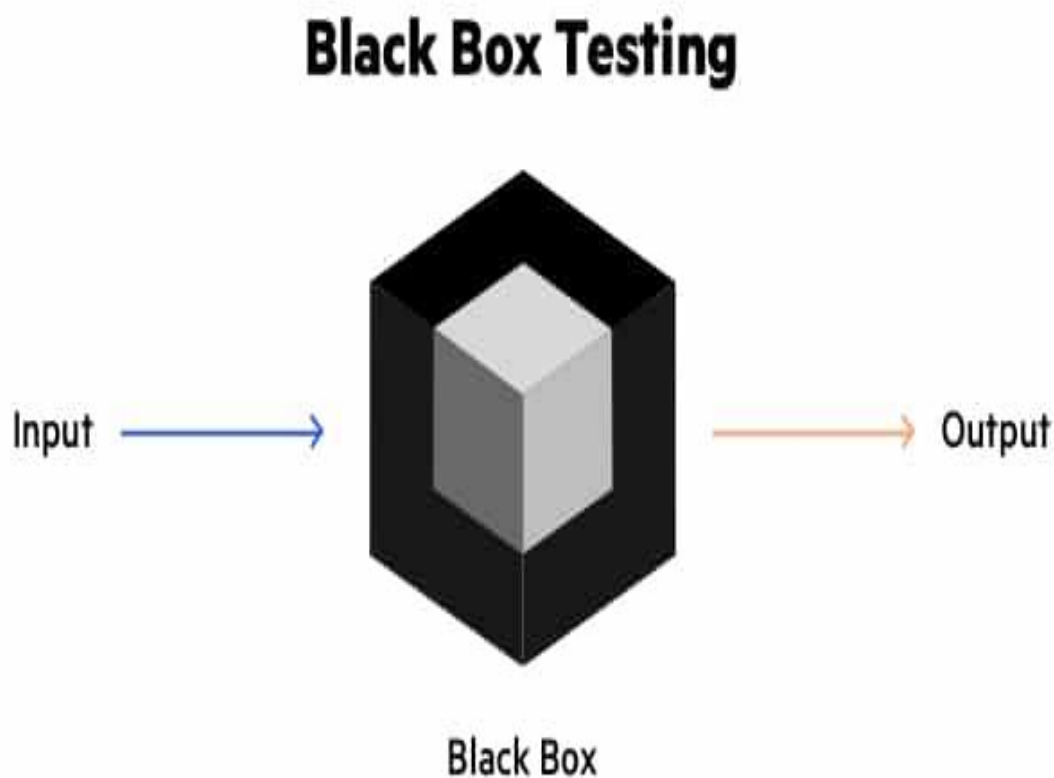


Figure 6.3.1.1: Black Box Testing

B. Functional/Acceptance Tests:

Tests from the customer's perspective that ensure that the system correctly implements a user story.

6.6 Validation Check:

6.6.1 Validation Check for Login:

If username and password is not correct then we select the login button show the same page. If username is valid password is not valid then show the same page. Or password is valid username is not valid then show the same page. If both username and password are valid then the form is submitting and go to home page according to their role. We also see the tables below.

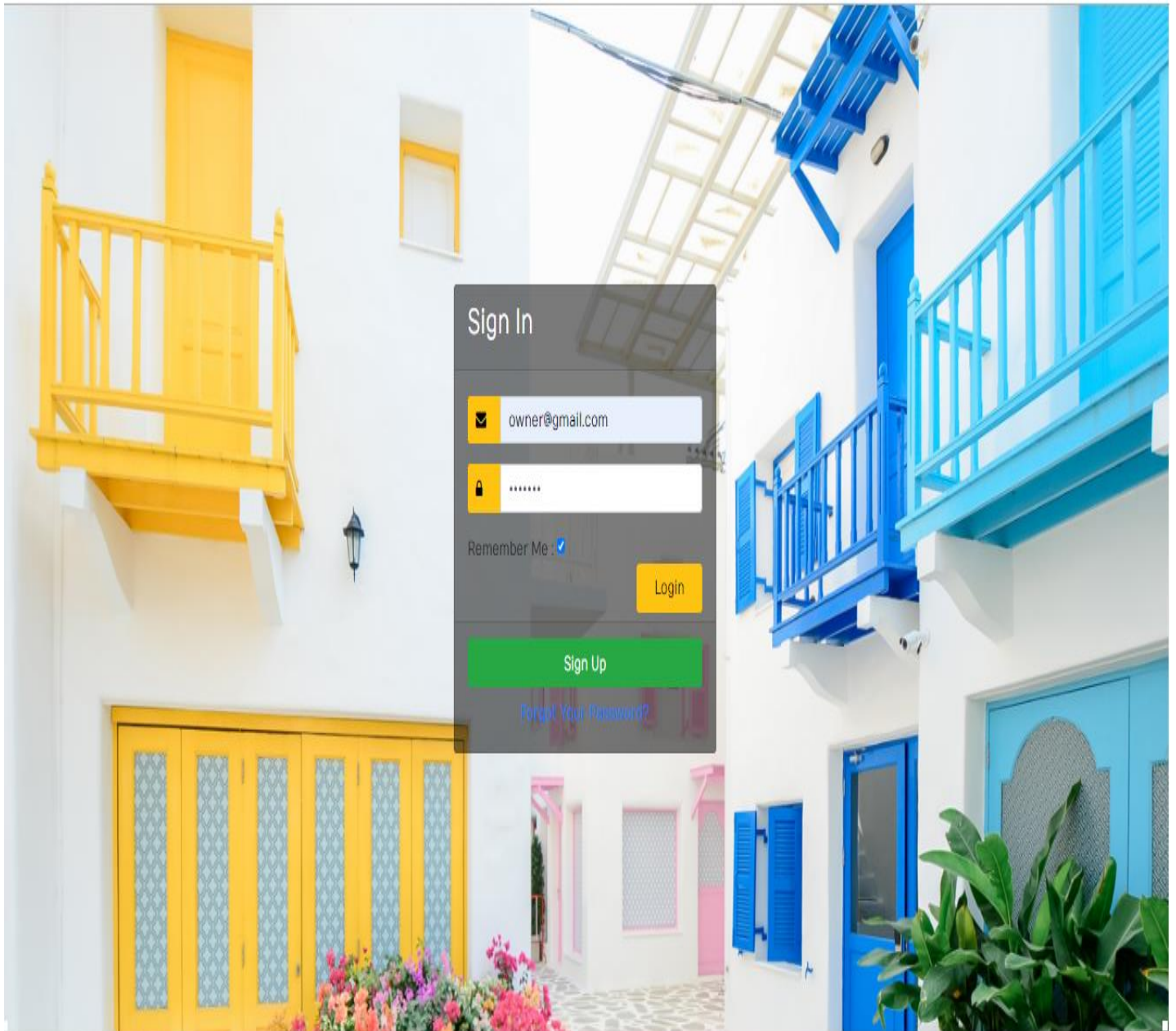


Figure 6.6.1: Sign in



Create User Account

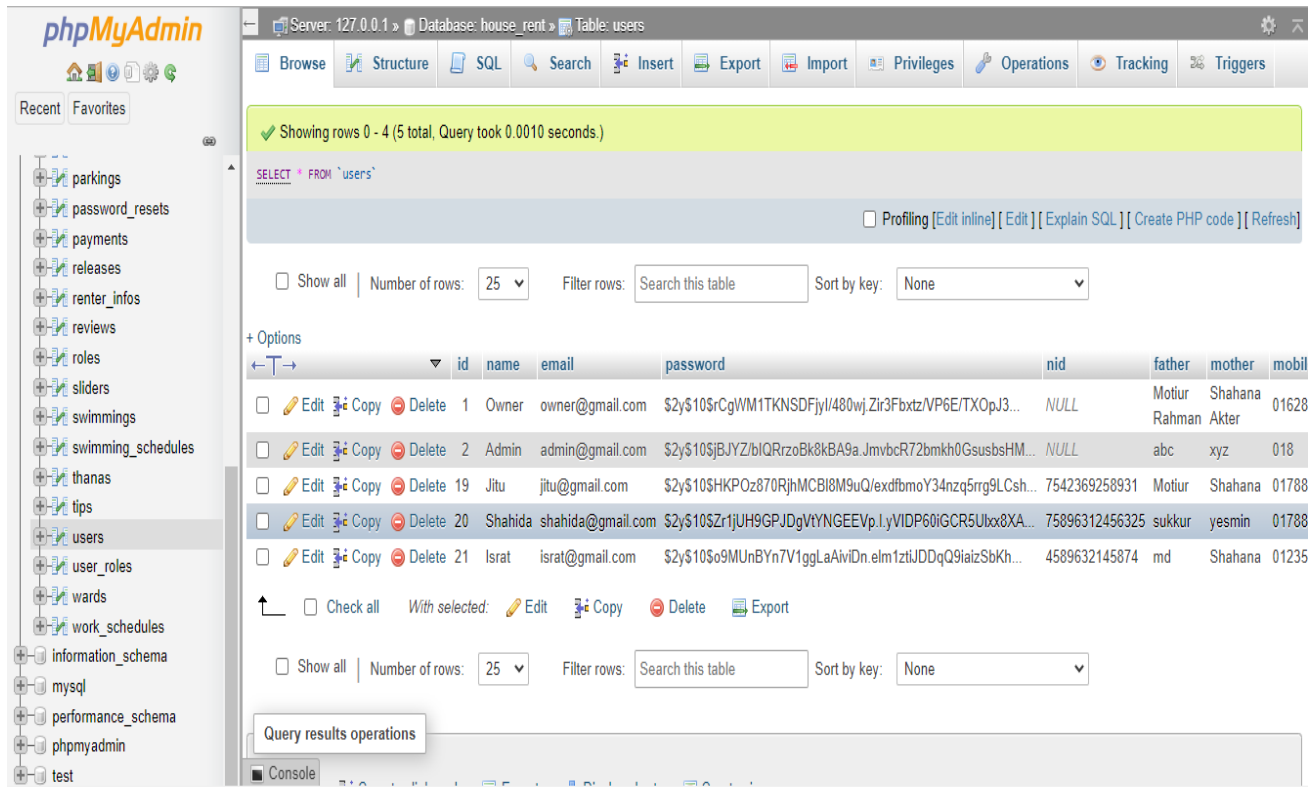
Name	Shahida	✓
Email	jtu@gmail.com	✓
Password	*****	
NID/Birth Certificate	7521369458796	✓
Father's Name	Motiur	✓
Mother's Name	Shahana	✓
Mobile No	01788359552	✓
	The mobile has already been taken.	
Blood Group	B+	✓
Image	Choose File No file chosen	✓
Permanent Address	Address	✓

Create Account

Figure 6.6.2: Create User Account

6.6.2 Login Validation Check Table:

If any username and password is not correct, there is no output only validator it's true then submit the form and login the site on their role:



The screenshot shows the phpMyAdmin interface for a database named 'house_rent'. The 'users' table is selected, and its structure and data are displayed. The table has columns: id, name, email, password, nid, father, mother, and mobil. The data is as follows:

id	name	email	password	nid	father	mother	mobil
1	Owner	owner@gmail.com	\$2y\$10\$rCgWM1TKNSDFjyl/480wjZir3Fbxtz/VP6E/TXOpJ3...	NULL	Motiuir Rahman	Shahana Akter	01628
2	Admin	admin@gmail.com	\$2y\$10\$BJYZ/bIQRzoBk8kBA9a.JmvbcR72bmkh0GsusbsHM...	NULL	abc	xyz	018
19	Jitu	jitu@gmail.com	\$2y\$10\$HKPOz870RjhMCBI8M9uQ/exdfbmoY34nzq5rrg9LCsh...	7542369258931	Motiuir Rahman	Shahana Akter	01788
20	Shahida	shahida@gmail.com	\$2y\$10\$Zr1jUH9GPJDgVtYNGEEVp.LyVIDP60IGCR5Ulx8XA...	75896312456325	sukkur	yesmin	01788
21	Israt	israt@gmail.com	\$2y\$10\$o9MUnBYn7V1ggLaAviDn.elm1ztIjDDqQ9iaizSbKh...	4589632145874	md	Shahana	01235

Figure 6.6.2 Login Validation Check Table

6.7 Summary:

This chapter focuses actual outcomes of the entries system. While testing we find out many defects that help to recovery errors and make our software more efficient and smoother. Any data such as owner, admin and user all are debugging step by step in this chapter.

CHAPTER-7

7 CONCLUSION

7.1 Introduction:

Our project is only a humble venture to satisfy the needs to manage one's project work. Several user-friendly coding has also adopted. This chapter mentions about the various issues which had been discussed in previous chapters. Our Apartment Management System is a Web-based application that helps the Apartment more unique and dynamic.

7.2 Project Outcomes:

We have mentioned all the features and functions to develop the system. We have discussed the development process of the system in previous chapters.

In this project, firstly owner can maintain the whole system and the modules will be provided for different system services.

- ☐ Secondly, the admin will also get access to log in for different modules.
- ☐ Thirdly owner and admin both can see the booking request and confirm the booking.
- ☐ Finally, renter can see the dashboard after accepting his request by owner or admin.

7.3 Future Works:

Some of the future work of the project below there:

- ☐ Android app will be added to the software.
- ☐ Cloud services will provide through the admin manager.
- ☐ In future we implement biometric and face recognition in our application so to reduce anti-social elements like robberies, murders and various other activities.
- ☐ Things like smart locks and smart thermostats, and in-home smart security systems are addressing the needs of renters. In the future, expect this technology to go beyond individual units, enabling residents to connect with the property in more ways.
- ☐ We will make this application for international purposes.

7.4 Summary:

In this project we highlighted the need of Apartment Management system which is a time saving process for the people who are living in the apartments. It also provides cardinal communication between the renters and owners relationships for better living and sharing information. Our project also helps to maintain security and providing scalability for n number of users to access the data without redundancy. It also provides flexibility for the tenants to share their documents, give complaints, pay their bills and various other issues etc. In this project efficiency of the application developed by our team initializes the few modules which may have few more modifications depending upon the user requirements. So, there will be scalability because this system is more secure as it is developed on web platform.

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